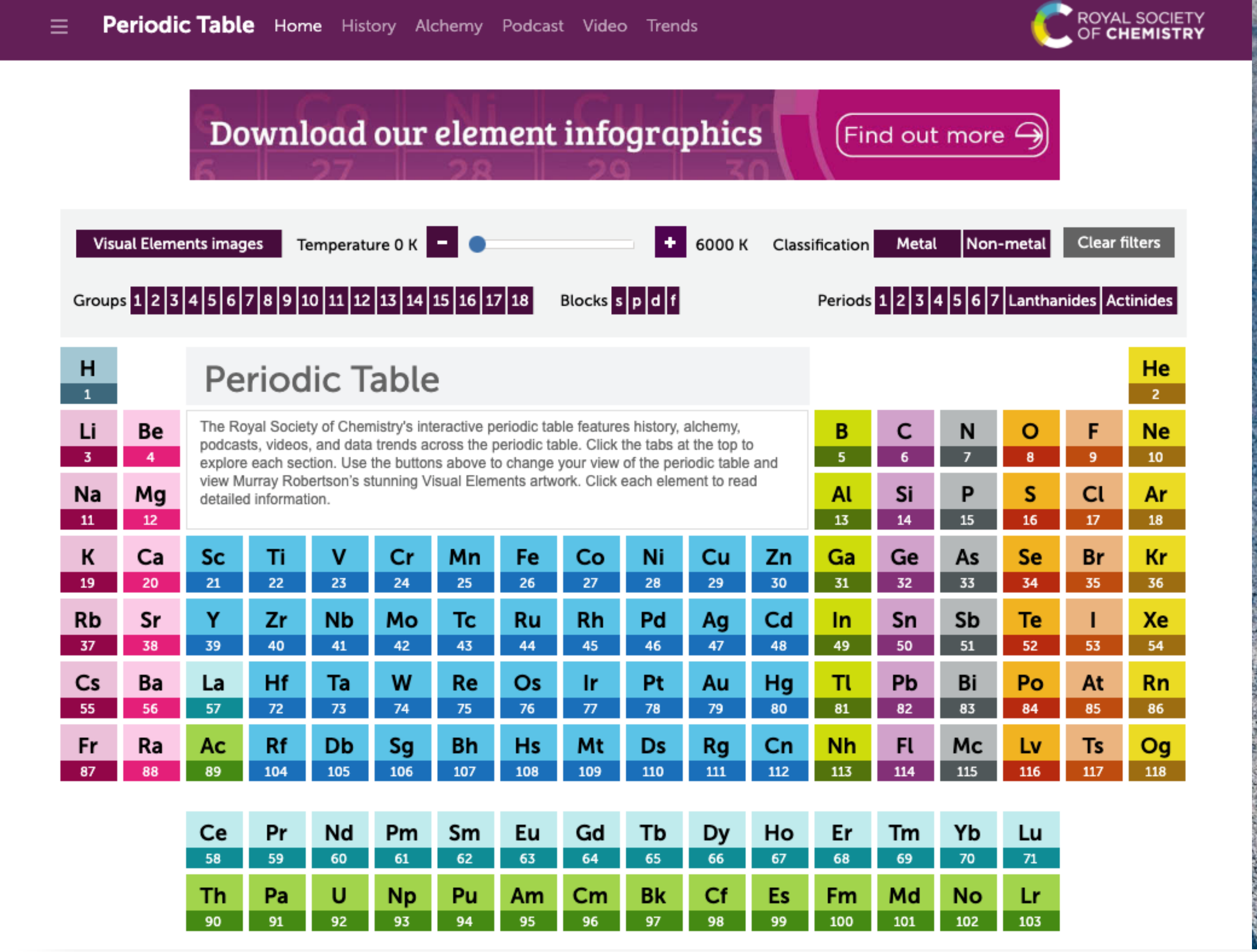


https://periodic-table.rsc.org/

CHAPTER 2: PERIODIC TABLE

PERIODIC TRENDS AND VARIATION OF PROPERTIES

CHAPTER OUTLINE

Classification of Elements

- Indicate period, group and block (s, p, d, f)
- Specify the position of metals, metalloids and non-metals in the Periodic Table
- Deduce the position of elements in the Periodic Table from its electronic configuration

Periodicity

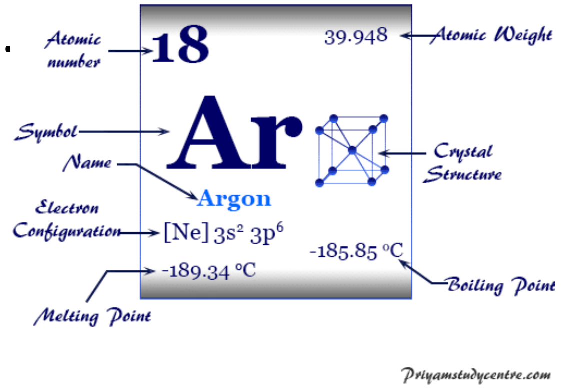
- Explain the variation in atomic and ionic radii
- Explain the radius of isoelectronic species
- Define the first and second ionisation species.
- Explain the increase in the successive ionisation energies of an element
- Explain the variations in the first ionisation energy:
  - across periods 2 and 3
  - down groups 1 and 2
- Deduce the electronic configuration of an element and its position in the Periodic Table based on successive ionisation energy data
- Define electron affinity and electronegativity
- Explain the variation in electronegativity of elements
  - across periods 2 and 3
  - down a group
- Describe the periodicity of elements across period 3 and down groups 1 and 17
- Describe and explain the acid-base character of oxides of elements in period 3

Skill

PERIODIC TRENDS

S, P, D, F

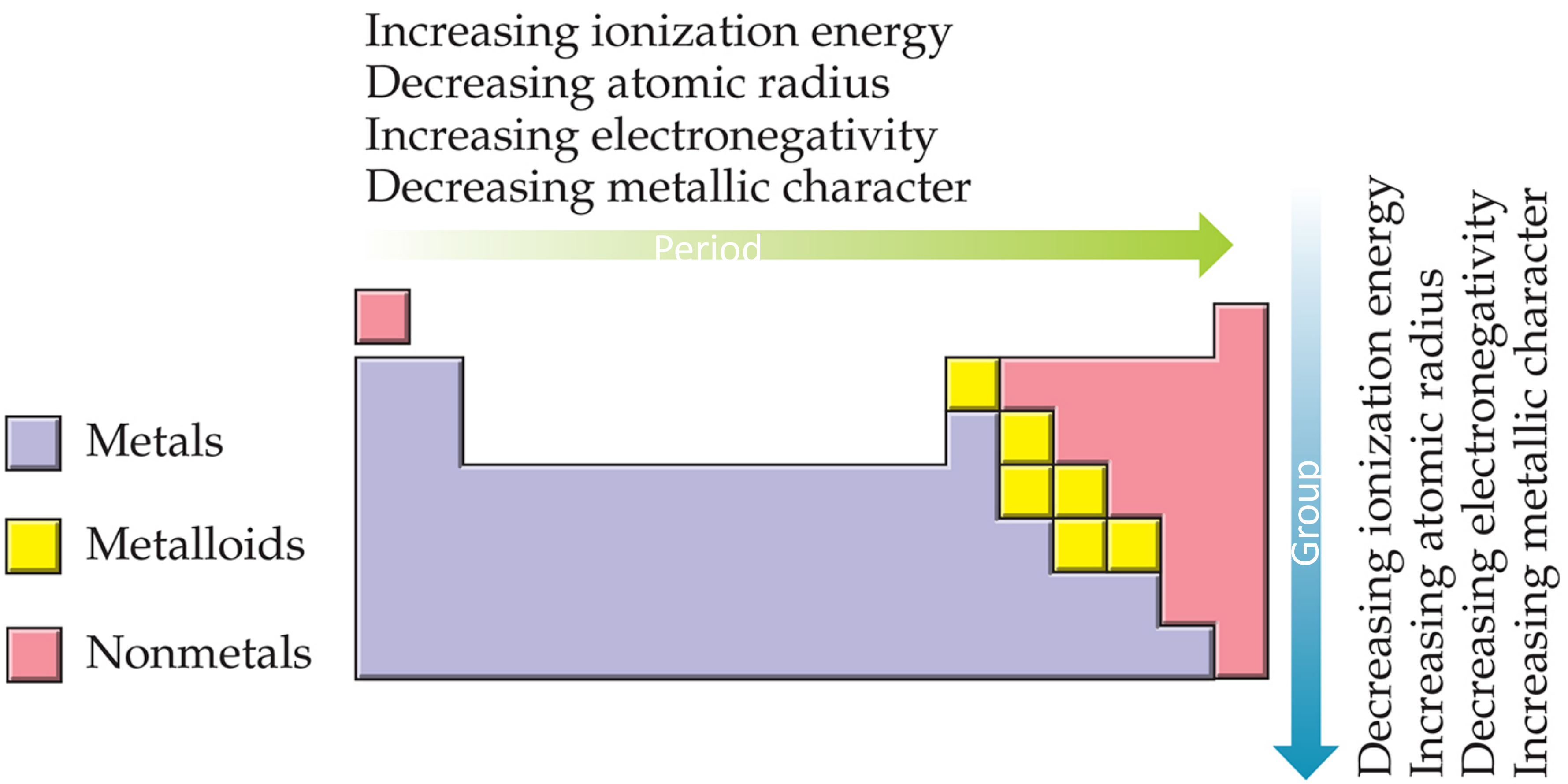
Portfolio



Trends

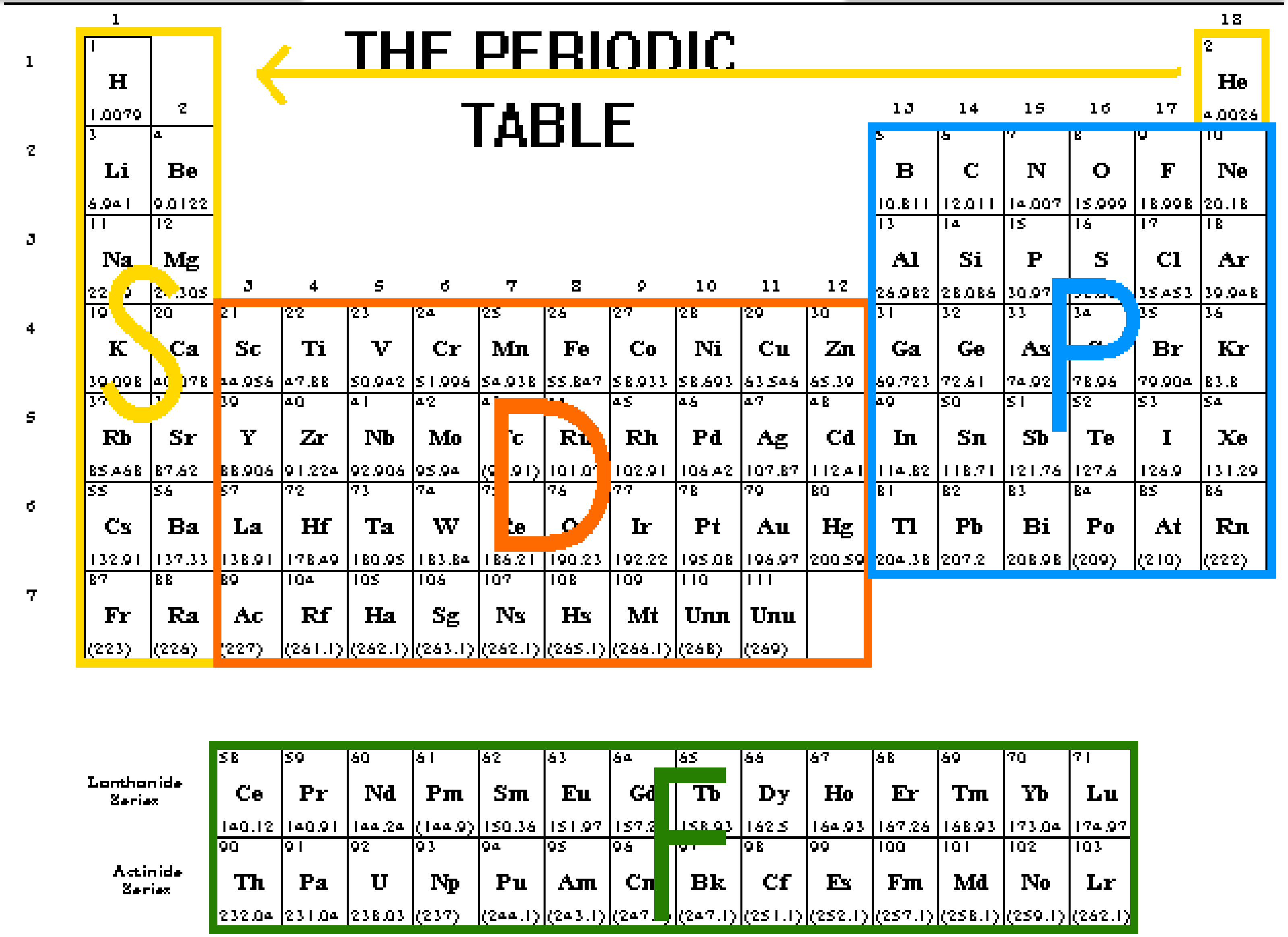
❖ increasing proton number; increasing atomic mass.

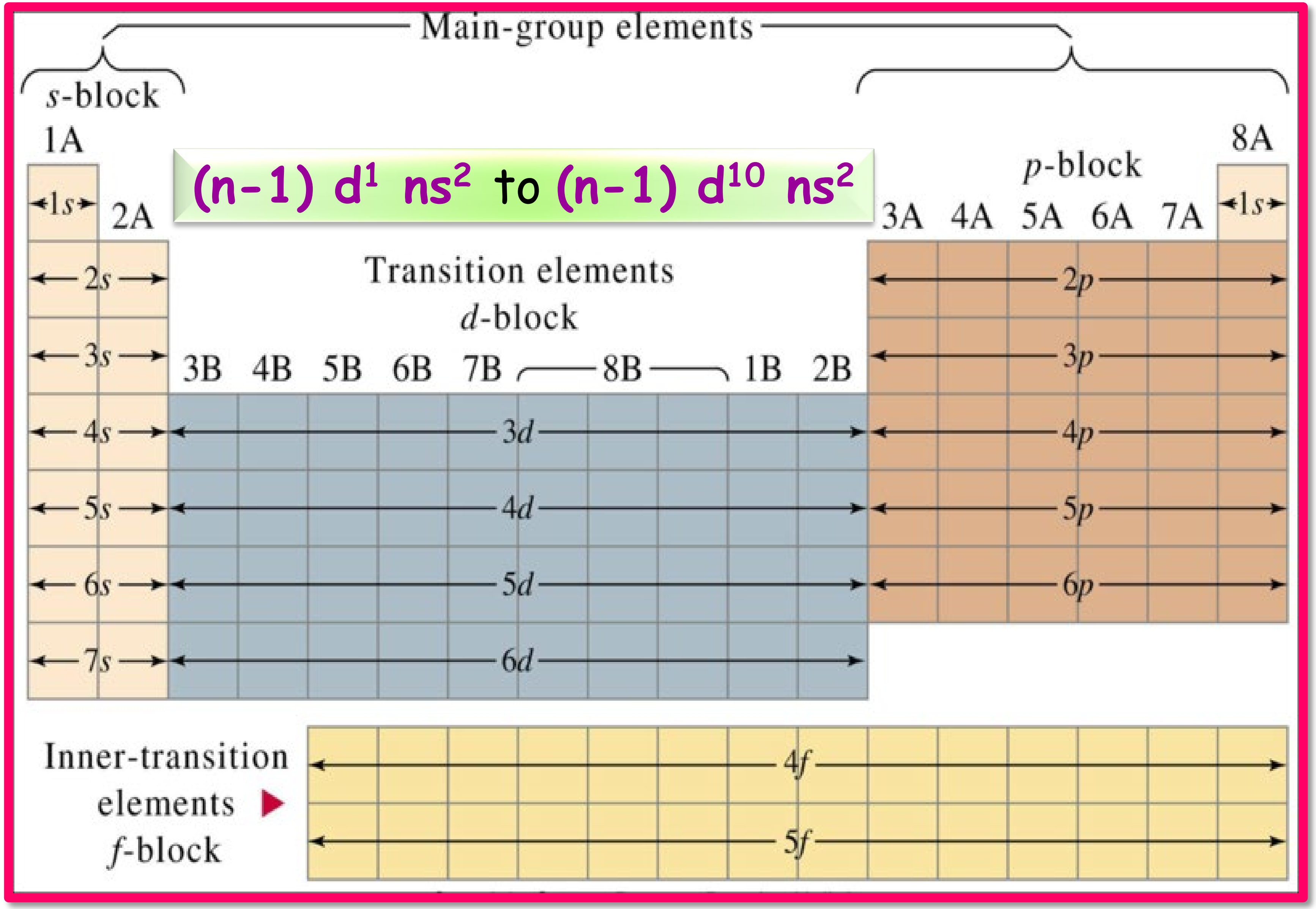
Period number = Principle quantum number



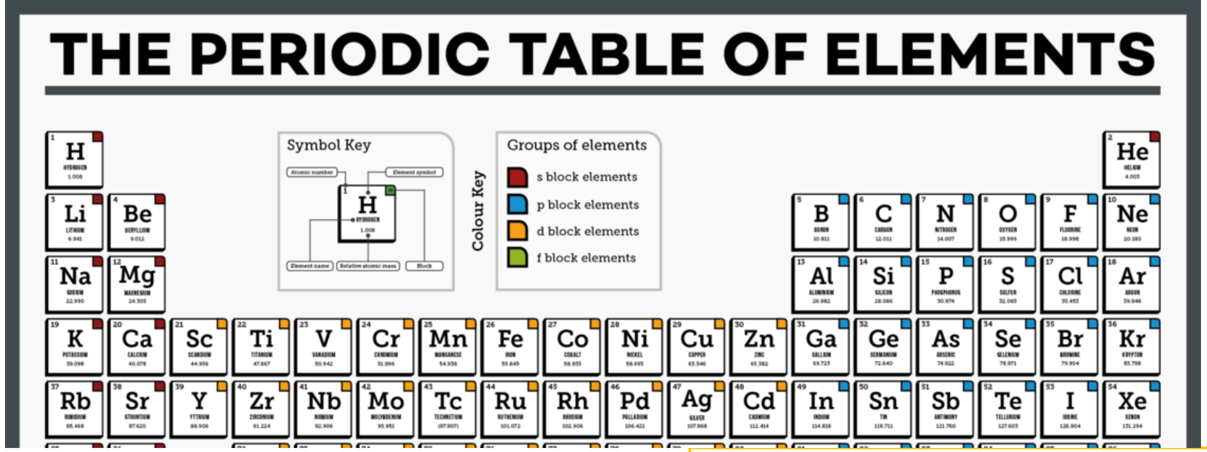
ns^1 to ns^2

$ns^2\ np^1$ to $ns^2\ np^6$





$(n-1)\ d^1\ ns^2$ to $(n-1)\ d^{10}\ ns^2$
Main-group elements
s-block
1A
2A
1s
2s
3s
4s
5s
6s
7s
p-block
8A
3A 4A 5A 6A 7A
2p
3p
4p
5p
6p
Transition elements
d-block
3B 4B 5B 6B 7B 8B 1B 2B
3d
4d
5d
6d
Inner-transition elements
f-block
4f
5f



Elements	Block	Period	Group	Num. of valence electron
$_{19}K$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ **Electron arrangement: 2.8.8.1**	**s**	**4**	**1**	**1**
$_{12}Mg$: $1s^2\ 2s^2\ 2p^6\ 3s^2$ **Electron arrangement: 2.8.2**	**s**	**3**	**2**	**2**
$_{15}P$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	**p**	**3**	**15**	**5**
$_{35}Br$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$ **Electron arrangement: 2.8.18.7**	**p**	**4**	**17**	**7**
$_{28}Ni$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$	d	4	10	10
$_{40}Zr$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^2$	**d**	**5**	**4**	**4**

PERIODIC TRENDS

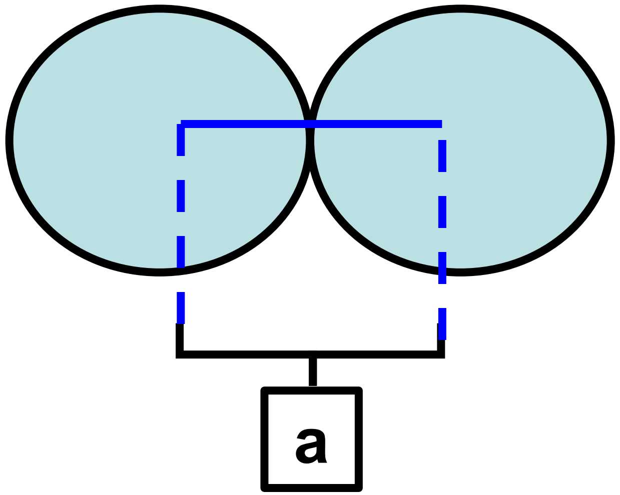
ATOM SIZE, ZEFF, IONIC RADIUS, ISOELECTRONIC, IONIZATION ENERGY, AFFINITY ENERGY, ELECTRONEGATIVITY

Portfolio

Periodic Trends In The Size Of Atom (Atomic Radii)

- The size /radius of atom is difficult to be defined exactly because the electron cloud has no clear boundary.

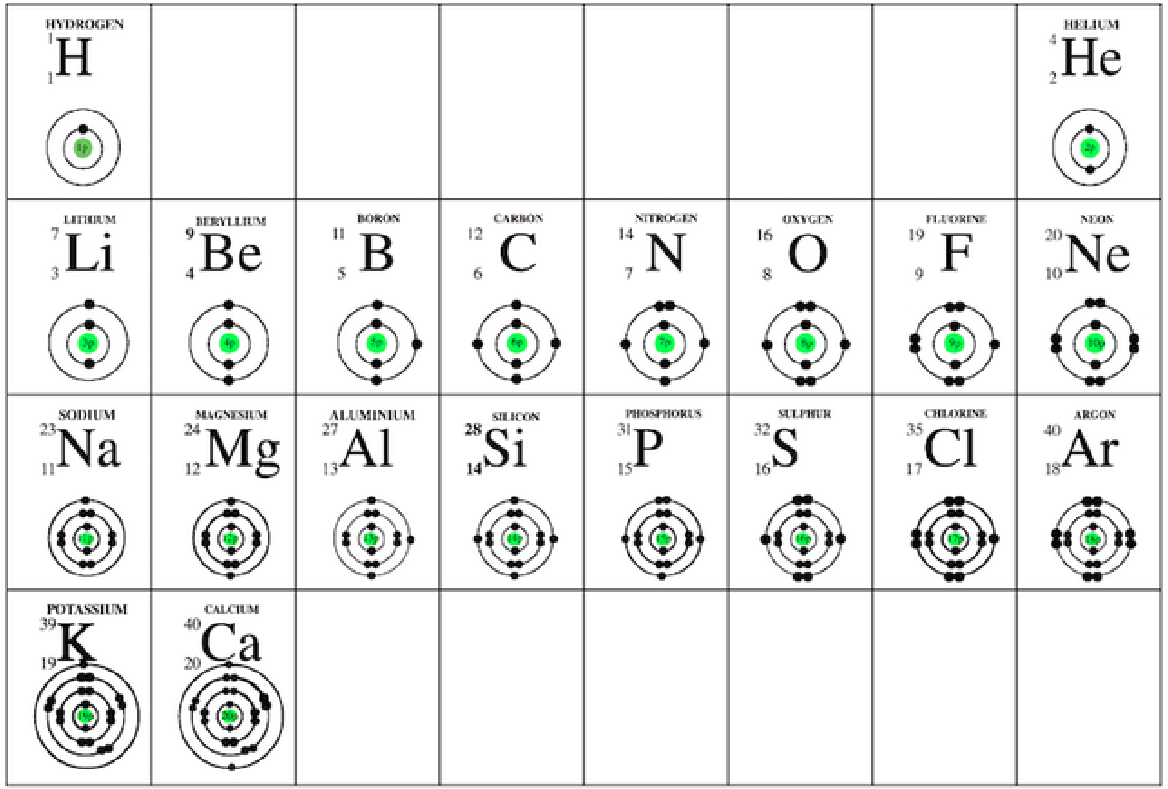
- **Atomic radius** is defined as half of the distance between the nuclei of two adjacent identical atom.



- Down the group, atomic radii increases.
- Across period, atomic radii decreases.

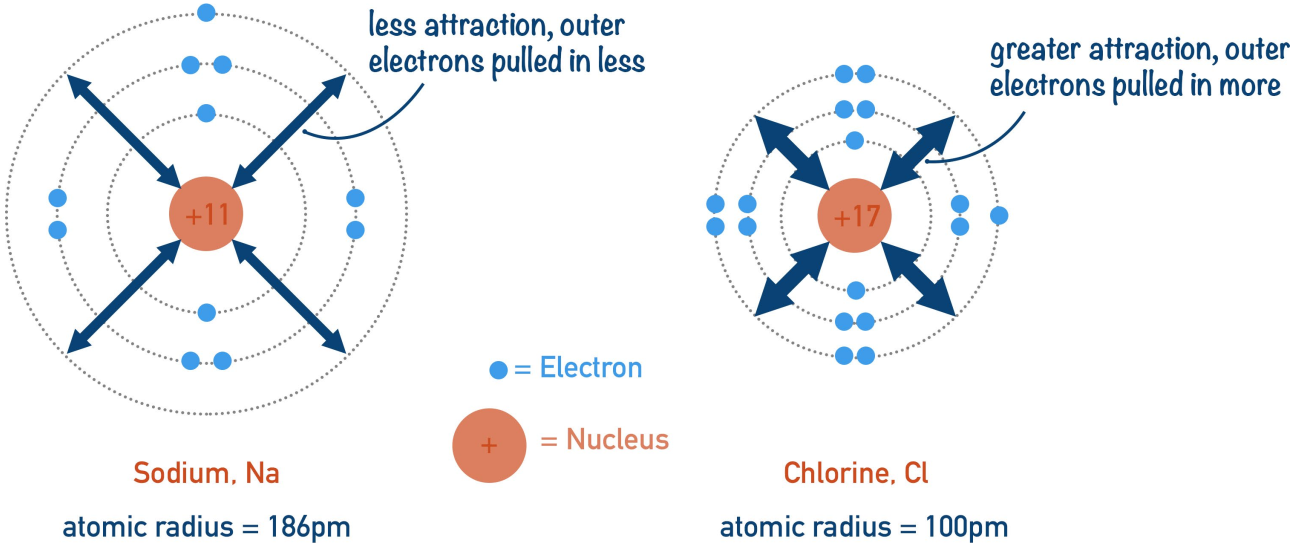
The changes of atomic radii in the Periodic Table is influenced by **TWO** factors:

- Effective nuclear charge experienced by the valence electrons.
- The principal quantum number, *n*, of the valence electrons



Effective Nuclear Charge (Z_{eff})

- Electrons around the nucleus experience different nucleus attraction.
- Those electrons **closer** to the **nucleus** experience a **greater attraction** than those that are farther away.
- The actual nuclear charge experienced by an electron is called the **effective nuclear charge, Z_{eff}**
- Effective nuclear charge increase, nucleus attraction stronger, atomic radii decrease.
- **Across** the **period,** the effective nuclear charge increases as proton number increase.
- As a result, the **attraction** between the nucleus and valence electrons become **stronger**, causing the **atomic radius** to **decrease**.



Effective Nuclear Charge

In general, the effective nuclear charge is given by

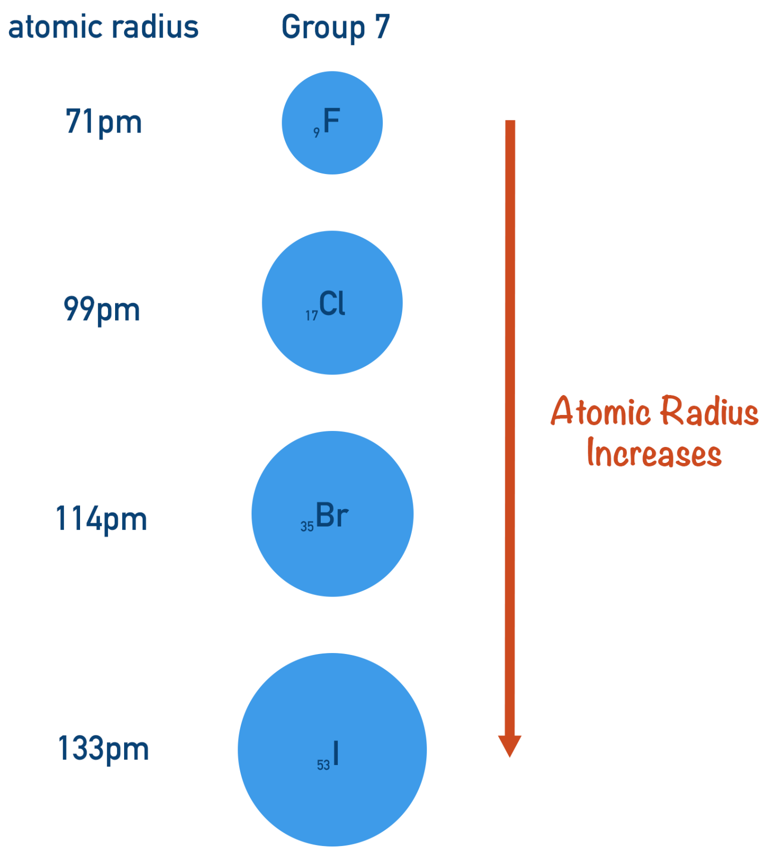
$$Z_{eff} = Z - \sigma$$

- Z is the nuclear charge or simply the number of protons in the nucleus.
- σ is the shielding constant.

Z_{eff} increases from left to right across a period; changes very little down a column.

Na: proton no 11, valence electron in 3s orbital
Inner electrone is 10 (in 1s,2s and 2p orbital shield)
Effective Nuclear Charge:
11-10 (in shield electron)= 1
Residual net charge felt by the valence electrons

Shielding Effect/Screening Effect



The shielding effect is caused by the mutual repulsion of electrons. Occurs between :

i) electrons of inner orbitals and electrons occupying the valence orbital

ii) electrons of same orbitals

- Extra inner shells effectively weaken the positive charge from the nucleus that reaches the outermost electrons; this is called inner electron shielding.

- Shielding effect causes the outermost electrons to be less attracted to nucleus.

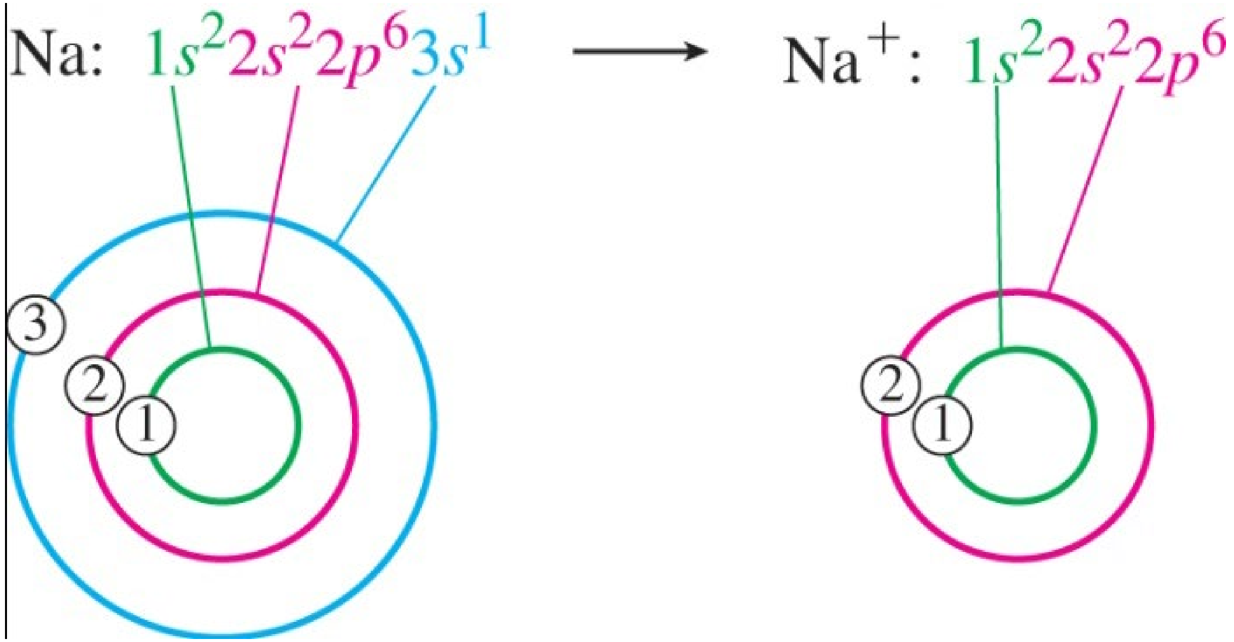
- Thus, the atomic size increases when shielding effect increases.

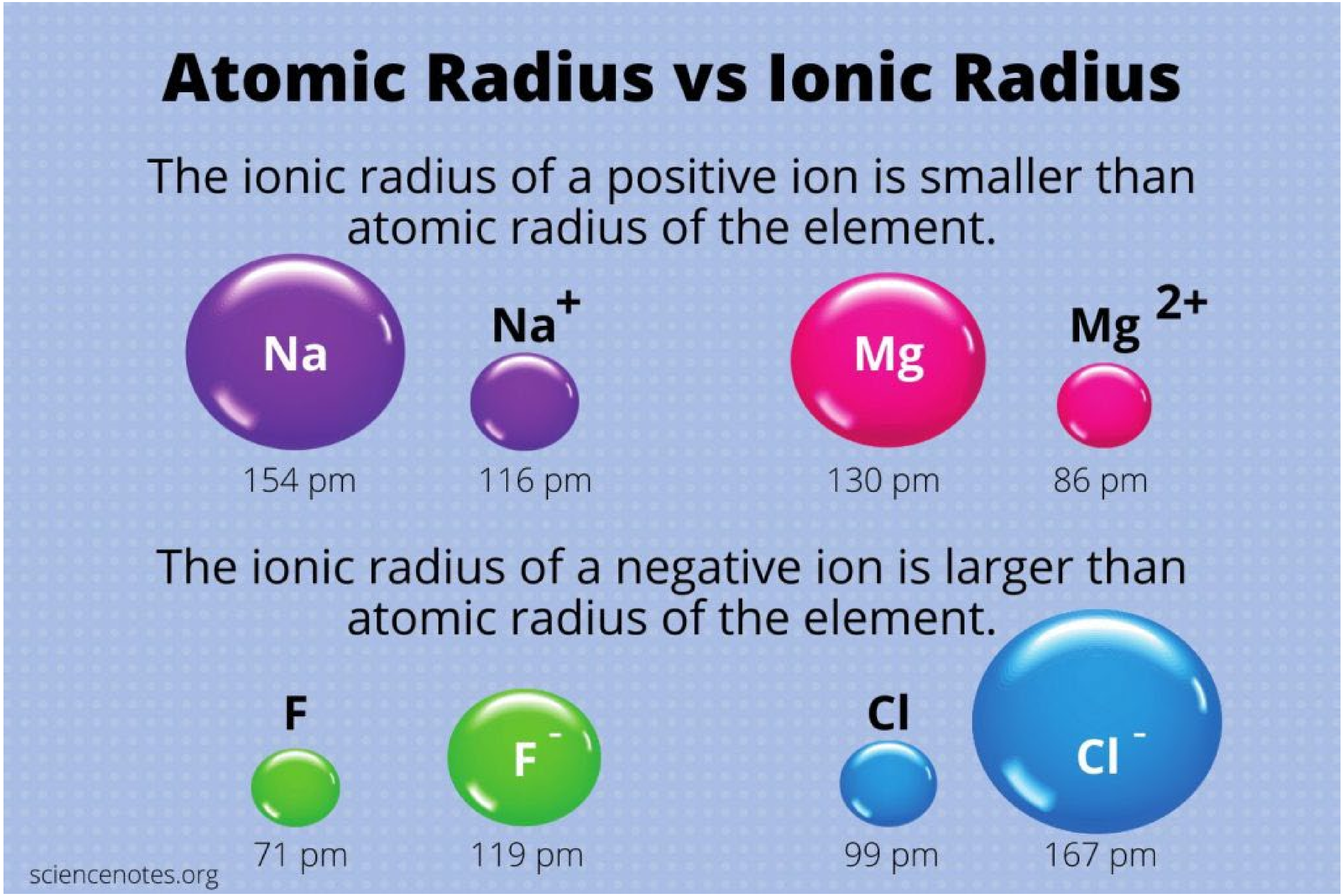
Ionic Radius

The ***ionic radius*** is the radius of a cation or an anion.

When an atom loses an electron to become a cation, its radius decreases due in part to a reduction in electron-electron repulsions in the valence shell.

A significant decrease in radius occurs when *all* of an atom's valence electrons are removed.



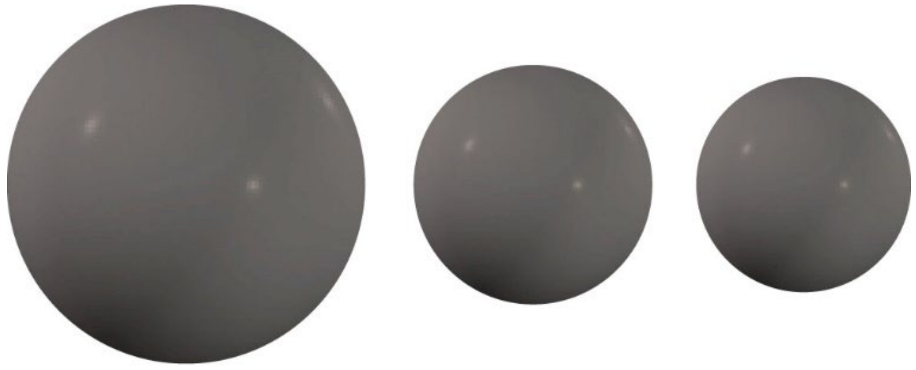

Atomic Radius vs Ionic Radius
The ionic radius of a positive ion is smaller than atomic radius of the element.
Na
154 pm
Na+
116 pm
Mg
130 pm
Mg 2+
86 pm
The ionic radius of a negative ion is larger than atomic radius of the element.
F
71 pm
F-
119 pm
Cl
99 pm
Cl-
167 pm
sciencenotes.org

Isoelectronic Species

- **Isoelectronic species** are groups of atoms and ions which have the same electronic configuration.

- Within isoelectronic species:

  - the more positive the charge, the smaller the species



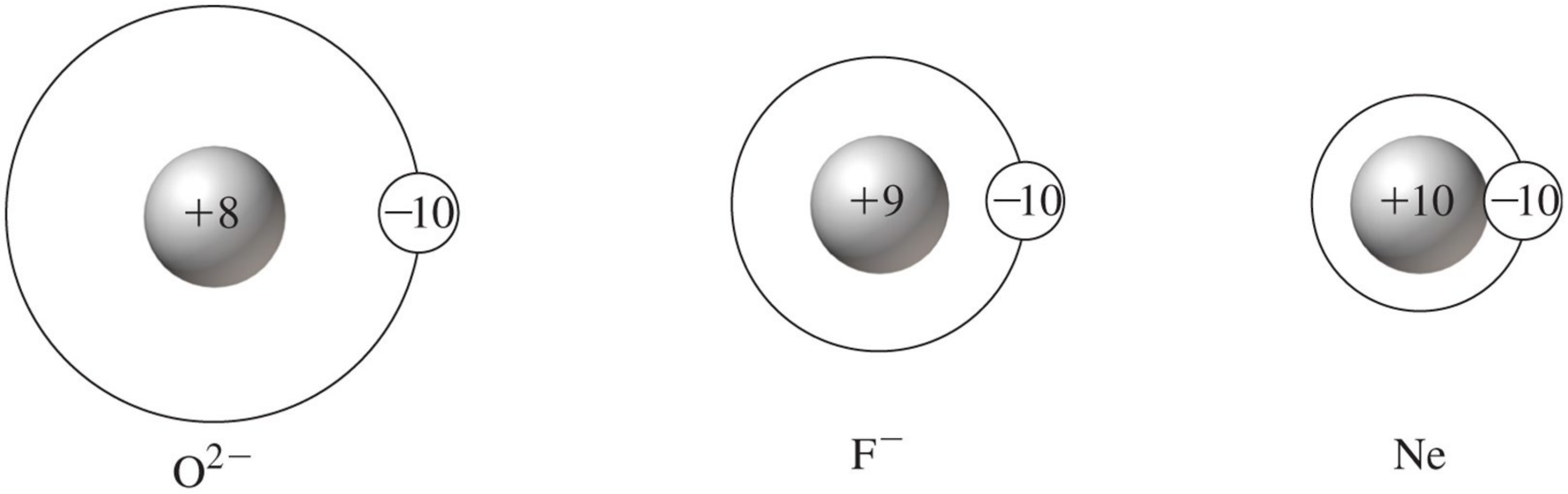
- Example :

  - Na^{+}, Mg^{2+}, $Al^{3+}$and Si^{4+} ions are isoelectronic
  - 10 electron where the electron configurations is $1s^22s^22p^6$.

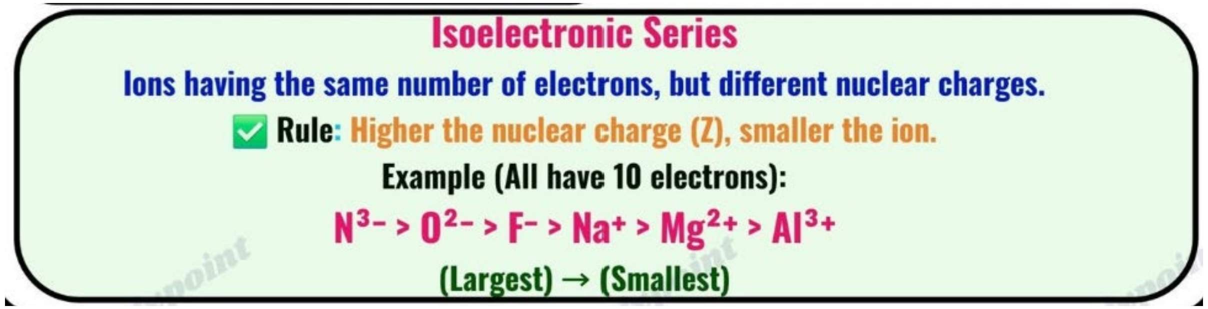
Isoelectronic Series

KEY IDEA: SAME ELECTRONS, DIFFERENT PROTONS (Z)

An ***isoelectronic series*** is a series of two or more species that have identical electron configurations, but different nuclear charges.



O^{2-} : $1s^2 2s^2 2p^6$ $\quad$ F^-: $1s^2 2s^2 2p^6$ $\quad$ Ne: $1s^2 2s^2 2p^6$



Isoelectronic Species With Electronic Configuration $1s^2\ 2s^2\ 2p^6$ (10 Electrons)

Species	Number of proton
Na^+	11
Mg^{2+}	12
Al^{3+}	13
Si^{4+}	14

- When proton number increase, effective nuclear charge increase.
- The attraction between nucleus and remaining electron increase.
- Therefore, the ionic radii decrease.
- The ionic radii of $Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

Isoelectronic Species With Electronic $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (18 Electrons)

Species	Number of proton
P^{3-}	15
S^{2-}	16
Cl^-	17

- When proton number increase, effective nuclear charge increase.
- The attraction between nucleus and remaining electron increase.
- Therefore, the ionic radii decrease.
- The ionic radii of $Cl^- < S^{2-} < P^{3-}$

Question

Consider the following isoelectronic species:

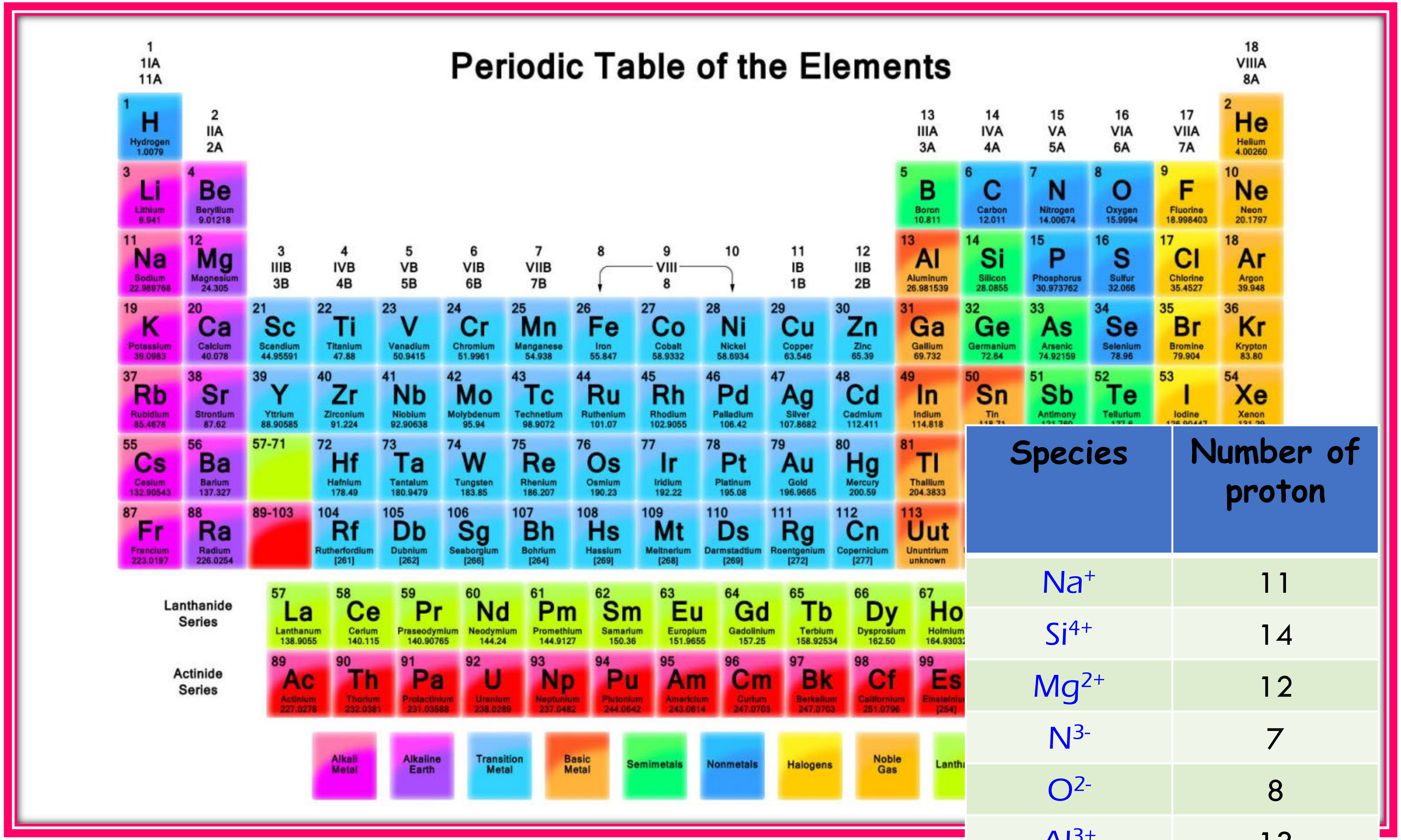
Na^{+}, Si^{4+}, Mg^{2+}, N^{3-}, O^{2-}, Al^{3+}, F^{-}

The electronic configuration of these isoelectronic is:

$1s^2\ 2s^2\ 2p^6$

Arrange those isoelectronic species in descending order the size.

Na^{+}, Si^{4+}, Mg^{2+}, N^{3-}, O^{2-}, Al^{3+}, F^{-}



Species	Number of proton
Na^{+}	11
Si^{4+}	14
Mg^{2+}	12
N^{3-}	7
O^{2-}	8
Al^{3+}	13
F^{-}	9

Question

Consider the following isoelectronic species:

$Na^+, Si^{4+}, Mg^{2+}, N^{3-}, O^{2-}, Al^{3+}, F^-$

The electronic configuration of these isoelectronic is:

$1s^2\ 2s^2\ 2p^6$

Arrange those isoelectronic species in descending order the size.

Species	Number of proton
Na^+	11
Si^{4+}	14
Mg^{2+}	12
N^{3-}	7
O^{2-}	8
Al^{3+}	13
F^-	9

Answer

$N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

TRENDS IN IONIZATION ENERGY

The ionization energy is the amount of energy it takes to detach one electron from a neutral atom

Or How much energy it takes to steal an electron!!

If its easy to steal an electron it has low ionisation energy

If its hard to steal an electron it has a high ionisation energy

The smaller the atom the harder to steal an electron

The larger the atom the easier it is to steal an electron

Ionization Energy

- **The ionization energy (IE)** is the minimum energy required to remove an electron from a gaseous atom in its ground state.

- The **first ionization energy** (IE_1) is the minimum energy required to remove the **first electron** from the atom in its ground state.

$$\text{Energy} + X(g) \rightarrow X^+(g) + e^- \qquad \Delta H = IE_1$$

Example:

$$Mg(g) \rightarrow Mg^+(g) + e^-$$

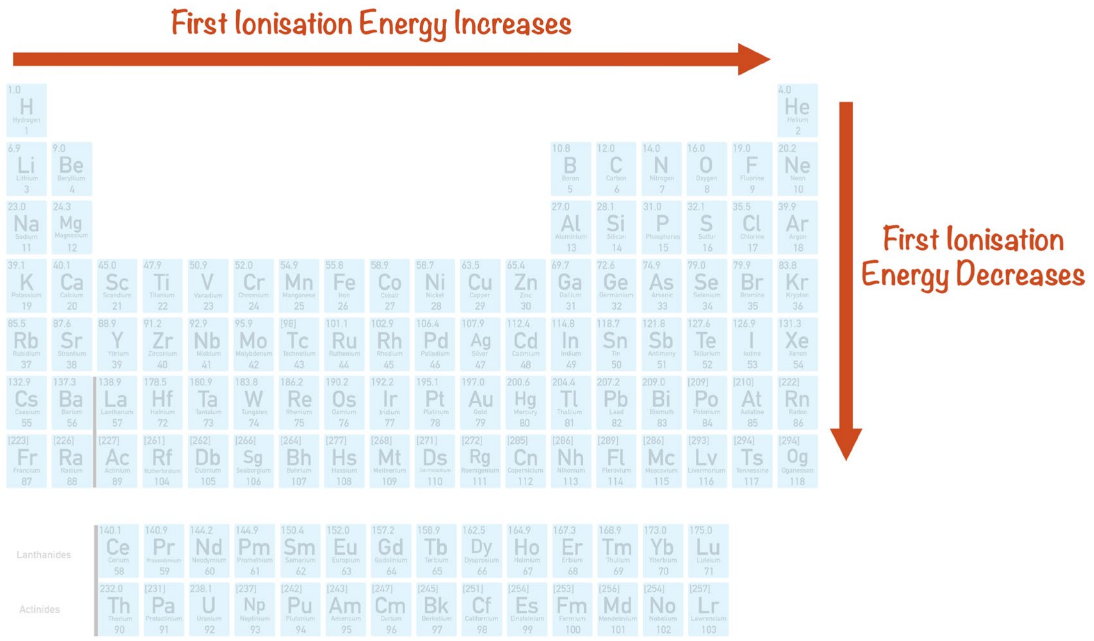
I_1 = first ionization energy

$$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$$

I_2 = second ionization energy

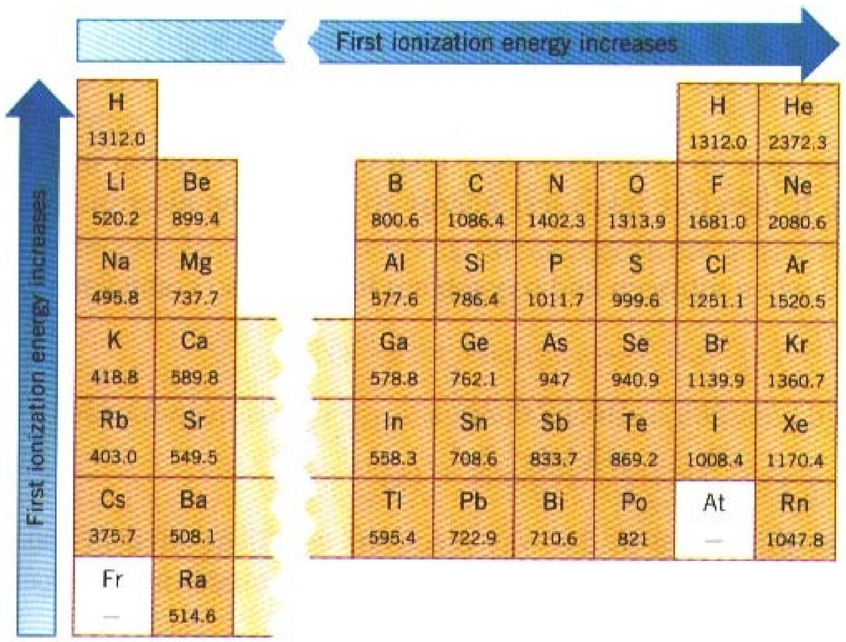
i) Ionization energy across a period :

- The effective nuclear charge increases, the atomic size decreases.
- Electrons are held tightly to the nucleus thus it is difficult to remove the first electron.
- Therefore, the first ionization energy is high.
- **The first ionization energy increases from left to right.**
- However, there are some irregularities in the trend.

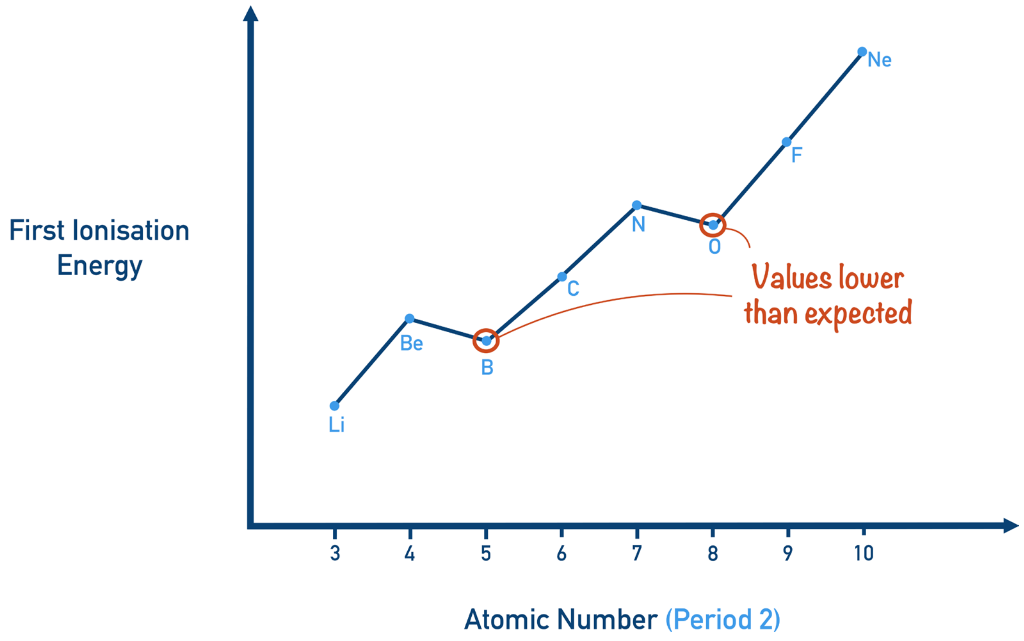


ii) Ionization energy going down the group

- Going **down the group**, the **atomic size increases** as the energy level, n increases.
- Therefore the outer **electrons** are farther from the nucleus and are **held less tightly** (weaker attraction) by the nucleus.
- Thus, it is **easy to remove** the **first electron.**
- Hence the Ionization Energy **decreases down the group.**



Anomalous cases in Group 2 & 13



The general trend across a period does have a couple of exceptions.

Boron a lower first ionisation energy than beryllium.

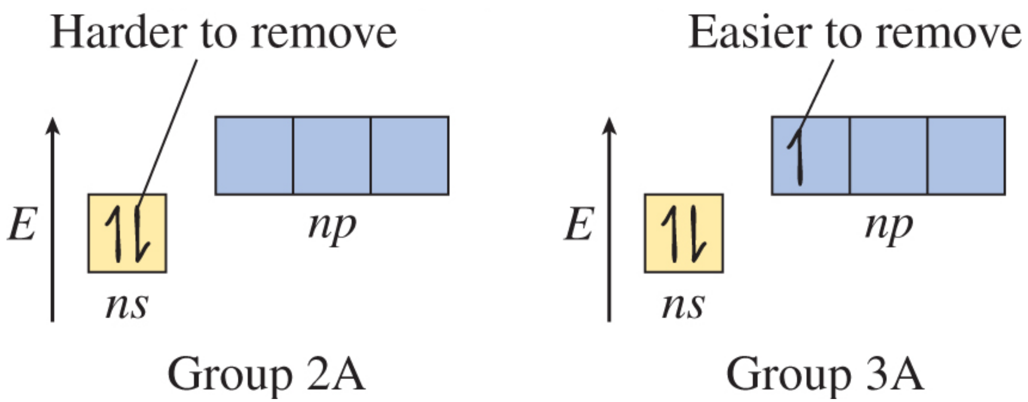
Oxygen has a lower first ionisation energy than nitrogen.

Beryllium electron configuration = $1s^2 2s^2$

Boron electron configuration = $1s^2 2s^2 2p^1$

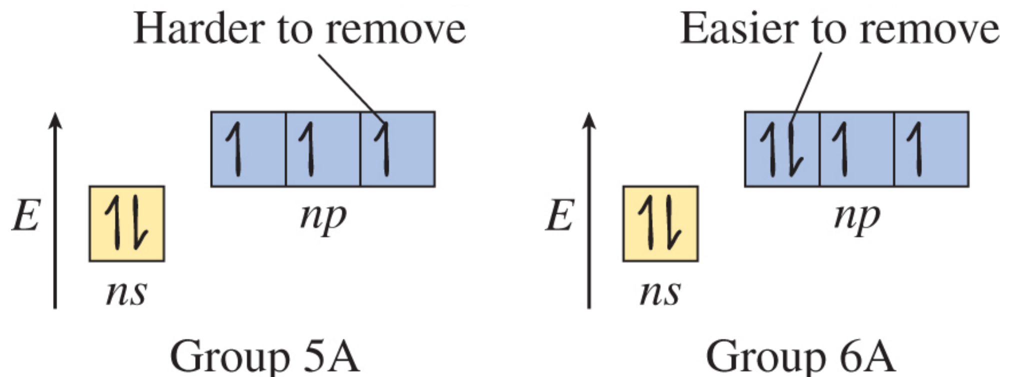
2p electron in boron is further from the nucleus, so it experiences less attraction and less energy is needed to remove it

- **Between group 2 and 13**
- $_5$**B** : $1s^2\ 2s^2\ 2\mathbf{p}^1$ in group 13 has a lower IE_1 than $_4$**Be**: $1s^2\ 2\mathbf{s}^2$ in group 2.
- **Be** loses a 2s electron while **B** loses a 2p electron.
- **Less** energy is needed to remove an electron from partially-filled **2p** orbital in B than to remove an electron from fully/completely filled **2s** orbital in Be.

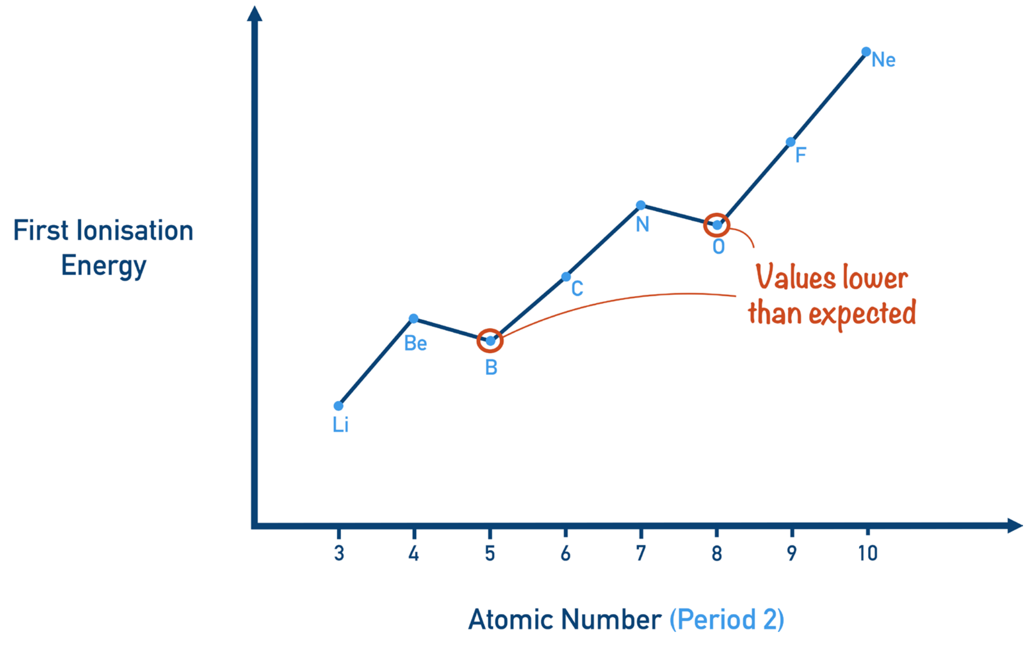


Between group 15 and 16

- **O** (group16) has **lower IE$_1$** than **N** (group 15)
- $_7$**N**:$1s^2\ 2s^2\ 2p^3$ (the **half-filled 2p** orbital)
- $_8$**O**:$1s^2\ 2s^2\ 2p^4$(the **partially-filled 2p** orbital)
- When N loses an electron it must come from the half-filled 2p orbital which is more stable than that of electron of the partially-filled orbital in O.
- As a result, **the first ionization energy of N is higher than of O.**



Anomalous cases in Group 15 & 16



The general trend across a period does have a couple of exceptions.

Oxygen has a lower first ionisation energy than nitrogen:

Nitrogen electron configuration = $1s^2 2s^2 2p^3$

Oxygen electron configuration = $1s^2 2s^2 2p^4$

★ *half-filled Nitrogen; harder to remove an electron*

- **Second ionization energy (IE_2)** is the minimum energy required to remove an electron from a positive gaseous ion.

$$\text{Energy} + X^{+}(g) \rightarrow X^{2+}(g) + e^{-}$$

- When an electron is removed from a neutral atom, the mutual repulsion among the remaining electrons decrease.
- Since the nuclear charge remain constant, the electron are held tightly to the nucleus.
- Therefore more energy is needed to remove another electron from the positively charged ion.

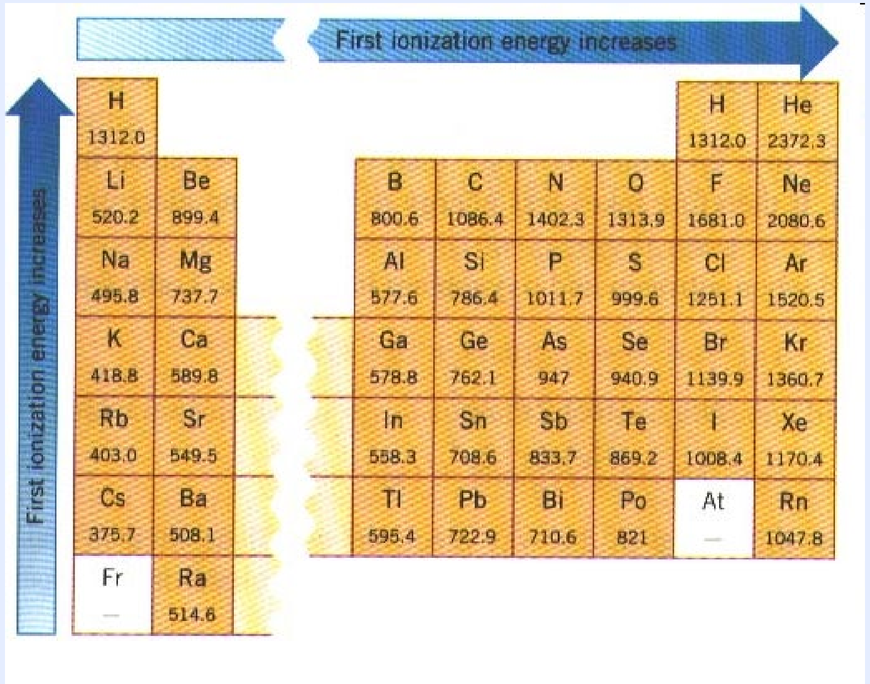
Increase E in the following order :

$IE_1 < IE_2 < IE_3 < IE_4 < \ldots..$

Worked Example

Would you expect Na or Mg to have the greater first ionization energy (IE_1)? Which should have the greater second ionization energy (IE_2)?

Strategy Consider effective nuclear charge and electron configuration to compare the ionization energies. Na has one valence electron and Mg has two.



Worked Example

Would you expect Na or Mg to have the greater first ionization energy (IE_1)? Which should have the greater second ionization energy (IE_2)?

Strategy Consider effective nuclear charge and electron configuration to compare the ionization energies. Na has one valence electron and Mg has two.

Solution

$IE_1(\text{Mg}) > IE_1(\text{Na})$ because Mg is to the right of Na in the periodic table (i.e., Mg has the greater Z_{eff}, so it is more difficult to remove its electron).

$IE_2(\text{Na}) > IE_2(\text{Mg})$ because the second ionization of Mg removes a valence electron, whereas the second ionization of Na removes a core electron.

Think About It The first ionization energies of Na and Mg are 496 and 738 kJ/mol, respectively. The second ionization energies of Na and Mg are 4562 and 1451 kJ/mol, respectively.

Section 6-3

Table 6.5 Successive Ionization Energies for the Period 2 Elements

Element	Valence Electrons	Ionization Energy (kJ/mol)*								
		1st	2nd	3rd	4th	5th	6th	7th	8th	9th
Li	1	520	7300							
Be	2	900	1760	14,850						
B	3	800	2430	3660	25,020					
C	4	1090	2350	4620	6220	37,830				
N	5	1400	2860	4580	7480	9440	53,270			
O	6	1310	3390	5300	7470	10,980	13,330	71,330		
F	7	1680	3370	6050	8410	11,020	15,160	17,870	92,040	
Ne	8	2080	3950	6120	9370	12,180	15,240	20,000	23,070	115,380

* mol is an abbreviation for mole, a quantity of matter.

Table 7.5 Successive Ionization Energies in Kilojoules per Mole for the Elements in Period 3

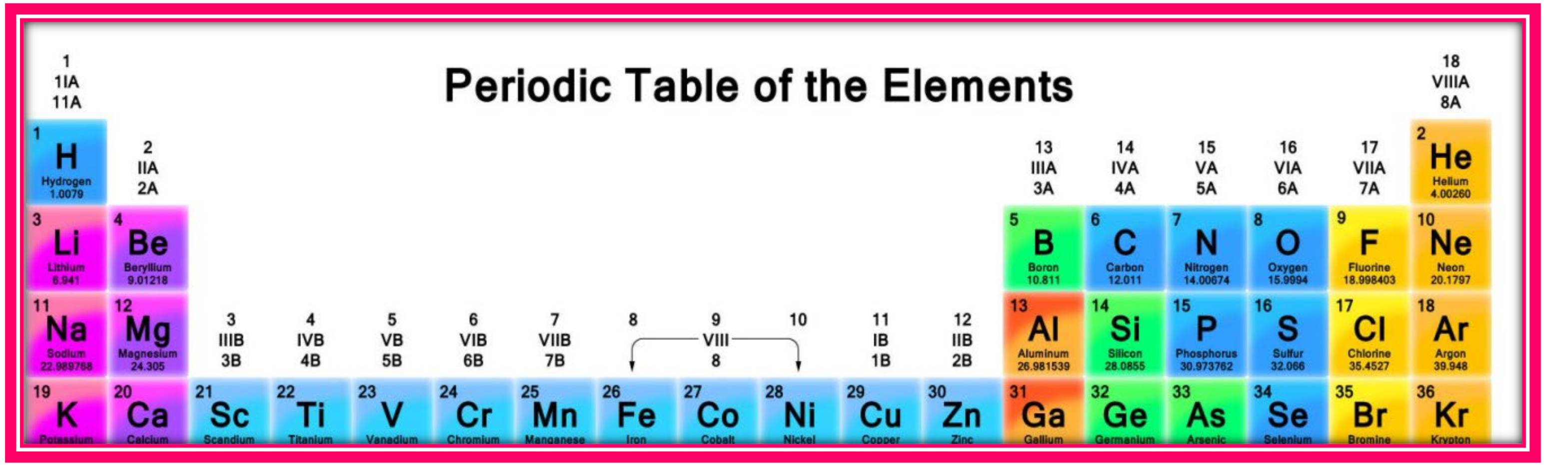
Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4560					
Mg	735	1445	7730	Core electrons*			
Al	580	1815	2740	11,600			
Si	780	1575	3220	4350	16,100		
P	1060	1890	2905	4950	6270	21,200	
S	1005	2260	3375	4565	6950	8490	27,000
Cl	1255	2295	3850	5160	6560	9360	11,000
Ar	1527	2665	3945	5770	7230	8780	12,000

*Note the large jump in ionization energy in going from removal of valence electrons to removal of core electrons.

General decrease

General increase

Question



Five successive ionization energies (kJmol^{-1}) of atom X is shown below:

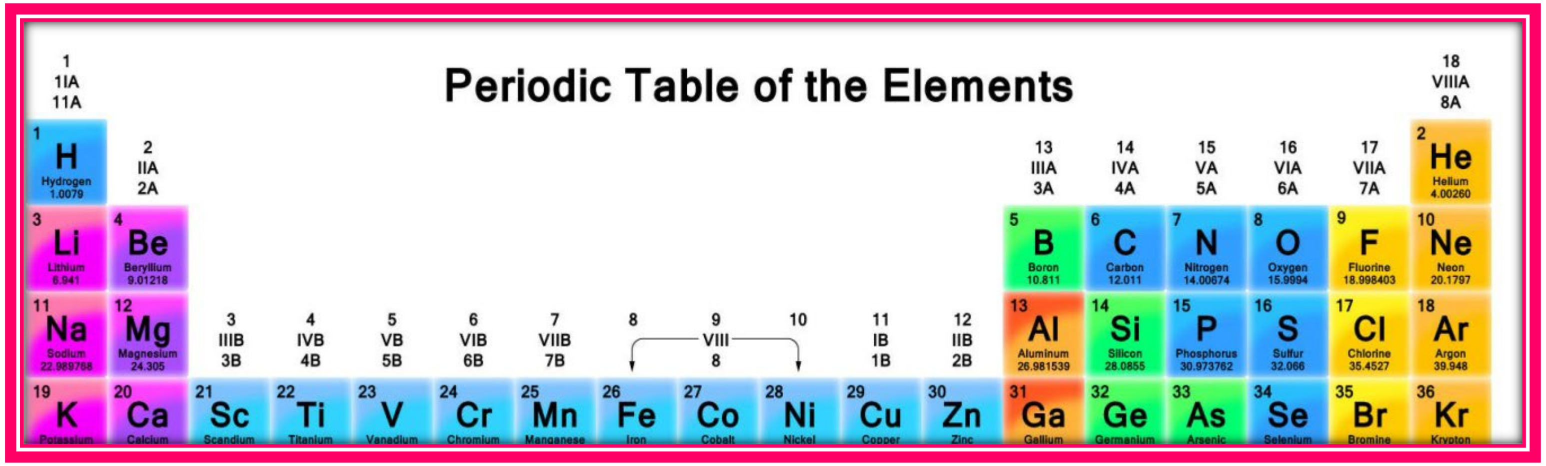
Ionization Energy	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7	IE_8
	999.6	2260	3380	4565	6996	8490	28080	31720

a) Deduce the electronic configuration of the valence electron of X.

b) Predict the group number in the periodic table to which X belongs.

c) Write the complete electronic configuration of X, if X is an element of the third period.

Question



Five successive ionization energies (kJmol^{-1}) of atom X is shown below:

Ionization Energy	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7	IE_8
	999.6	2260	3380	4565	6996	8490	28080	31720

a) Deduce the electronic configuration of the valence electron of X. $ns^2\ np^4$

b) Predict the group number in the periodic table to which X belongs. Group 16

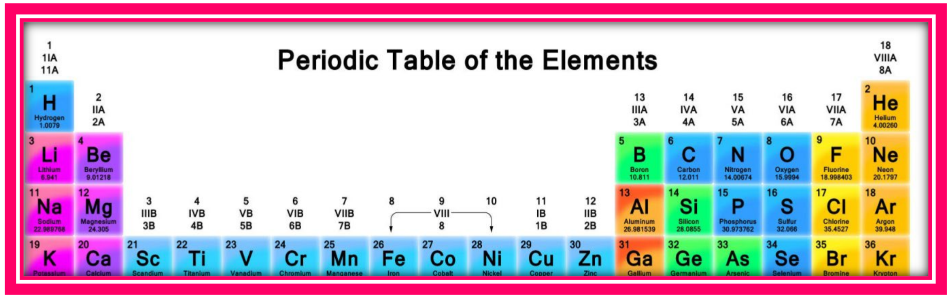
c) Write the complete electronic configuration of X, if X is an element of the third period. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

Trends in the Electron Affinity, EA

Electron affinity : Electron affinity : the change in energy (in kJ/mole) of a neutral atom (in the gaseous phase) when an electron is added to the atom to form a negative ion.

Which one is higher or lower value Ea?

We said:



Across Period; Love electron: easy to fill in outer shell, energy released, more exothermic, EA value more negative!

- First Electron Affinity (negative energy because energy released):

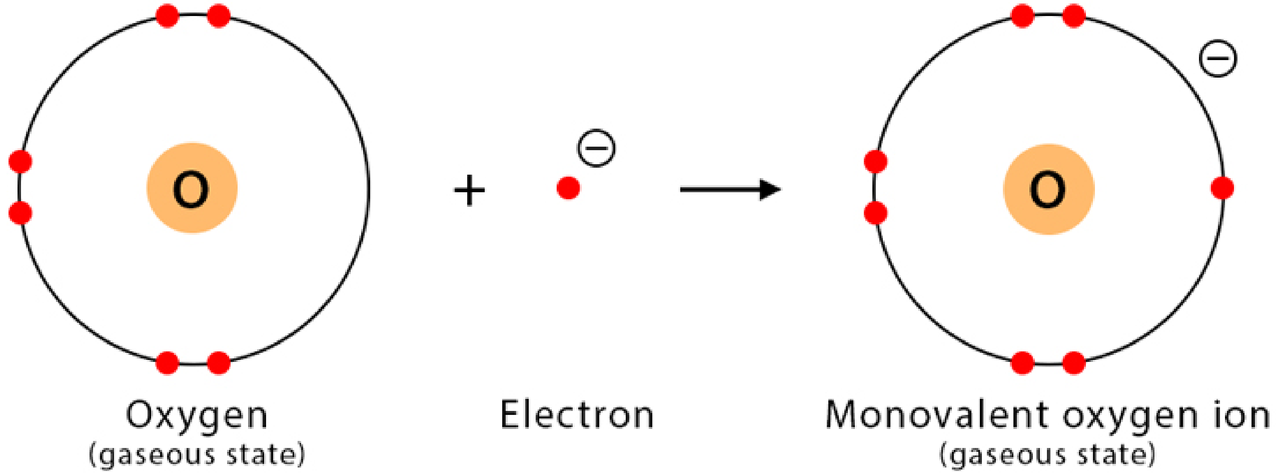
$$X(g) + e^- \rightarrow X^-(g) \quad \Delta EA = -ve\ value(exothermic)$$

- Second Electron Affinity (positive energy because energy needed is more than gained (repulsion)):

$$X^-(g) + e^- \rightarrow X^{-2}(g)\ \Delta EA = +ve\ value(endothermic)$$

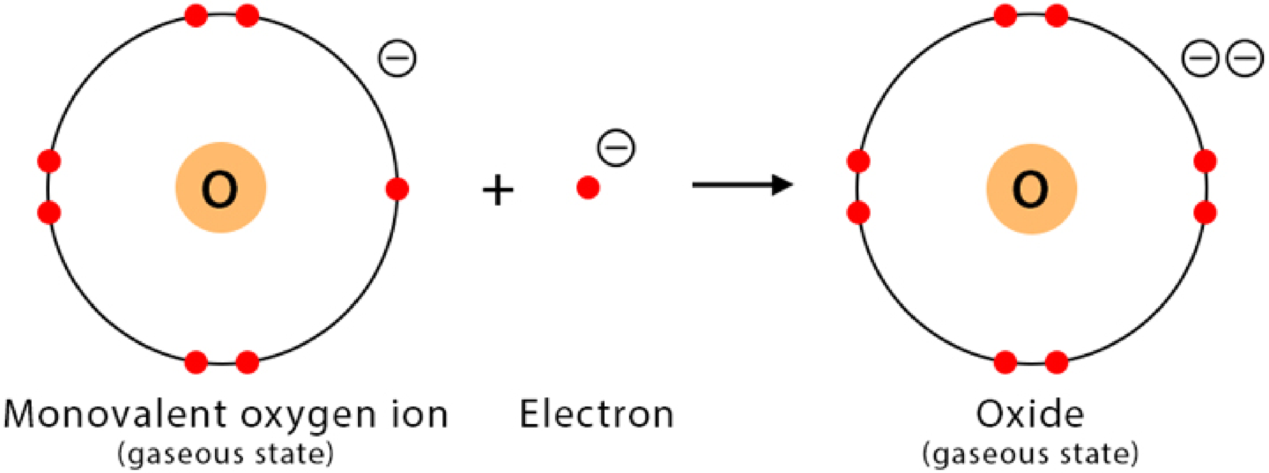
Second Electron Affinity

First Electron Affinity



Energy released: EA_1 = -142 kJ/mol

Second Electron Affinity



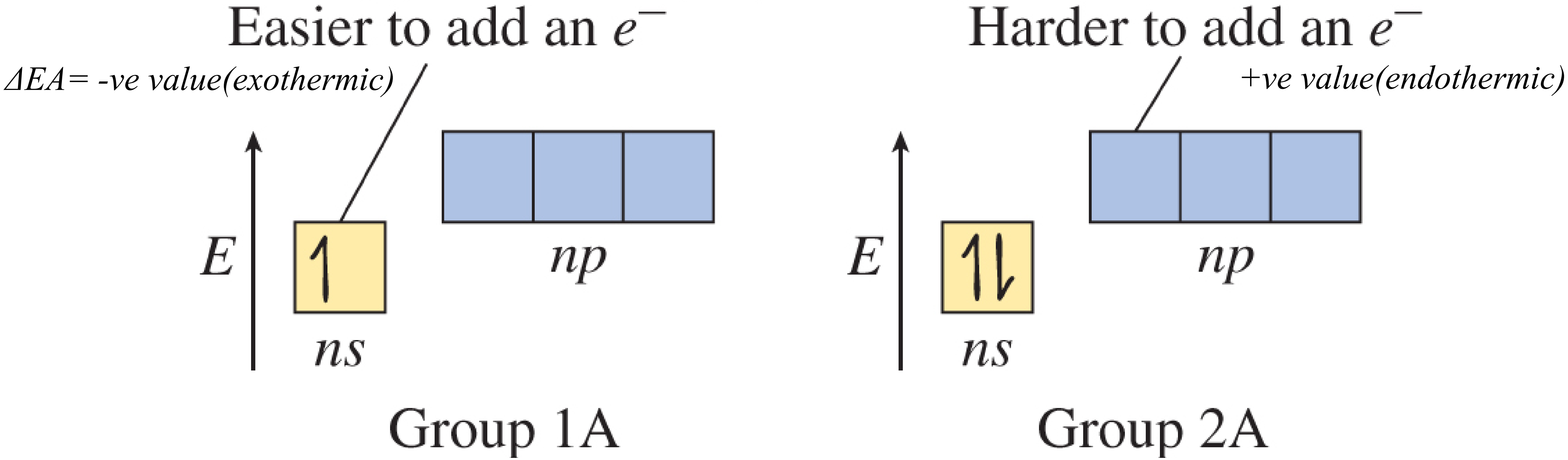
Energy absorbed: EA_2 = 844 kJ/mol

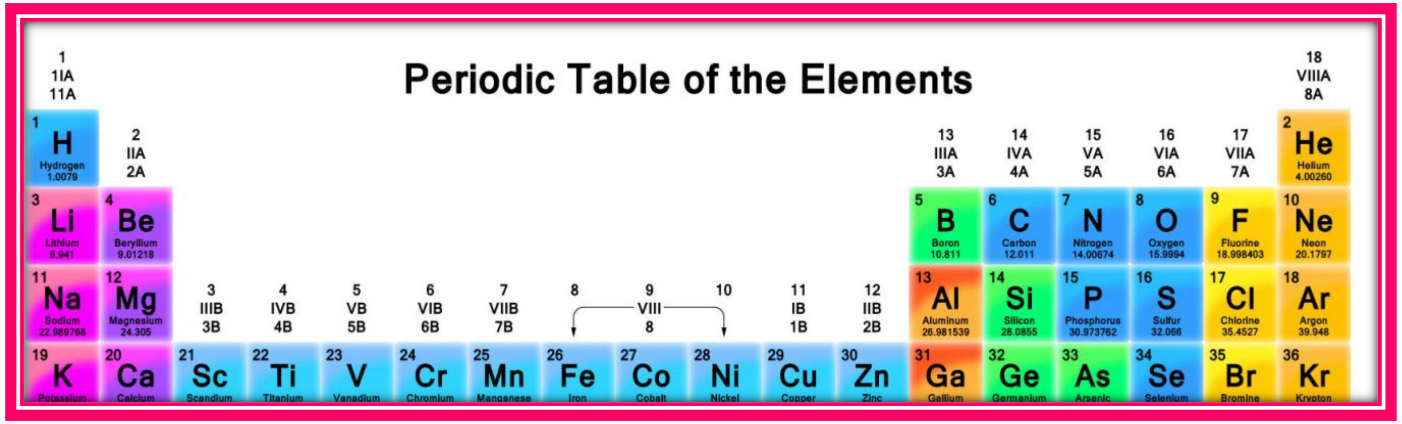
Anomalies Trends in the Electron Affinity, EA

Electron affinity : energy release per mole(in kJ/mole) (in the gaseous phase)

Across period: EA increase with more negative value due to strong proton number, but anomalies group 2 and group 13:

It is easier to add an electron to an *s* orbital than to add one to a *p* orbital with the same principal quantum number.

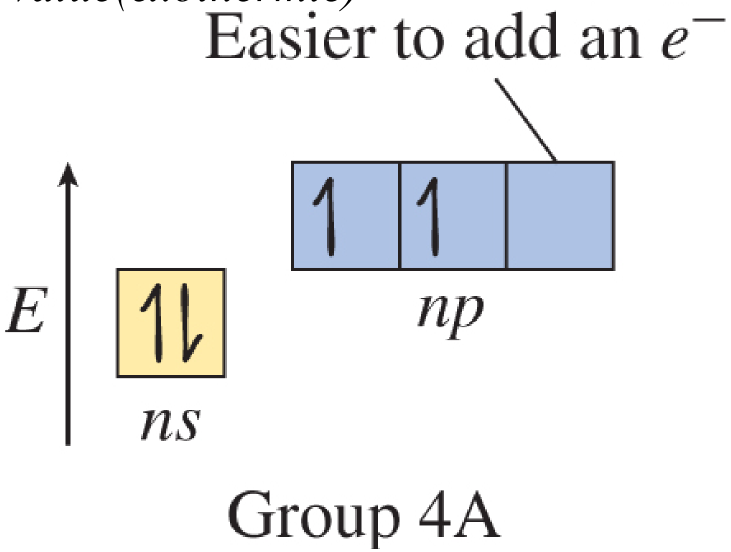




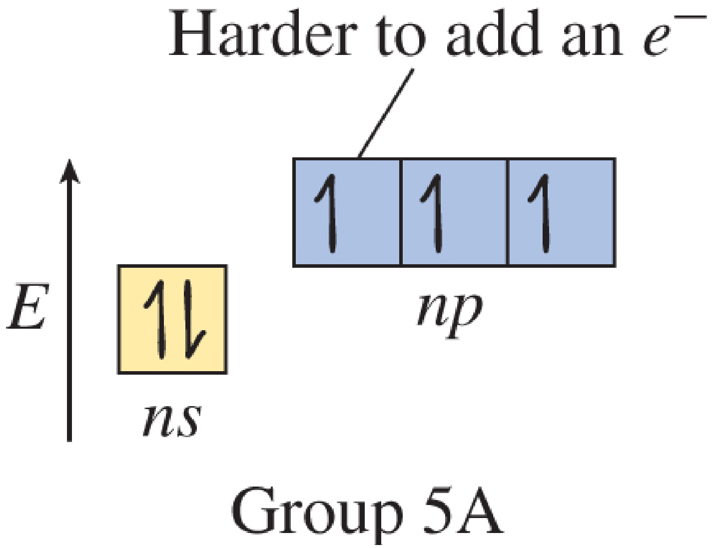
Across period: EA increase with more negative value due to strong proton number, but anomalies group 15 and group 16:

Within a *p* subshell, it is easier to add an electron to an empty orbital than to add one to an orbital that already contains an electron.

ΔEA= -ve value(exothermic)

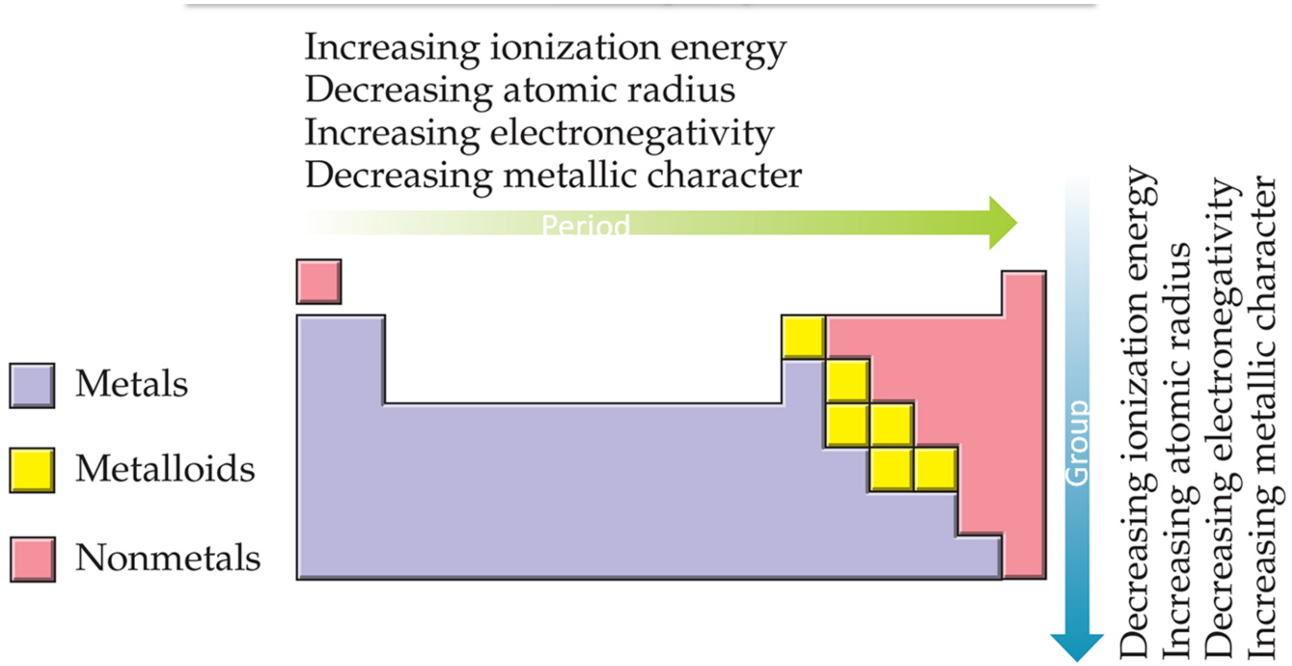


+ve value(endothermic)

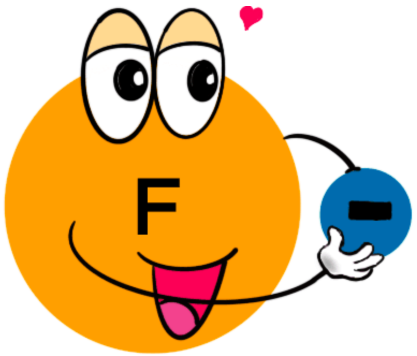


Why Do Metals Have A Low Electron Affinity?

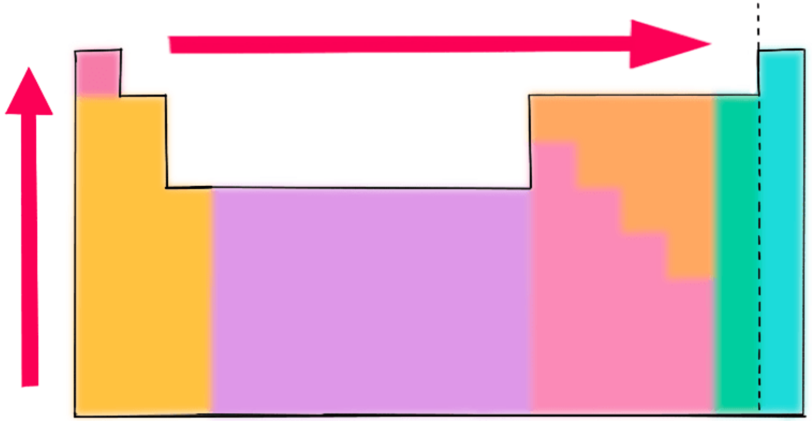
Metals have a low electron affinity (a less likely chance to gain electrons) because they want to give up their valence electrons rather than gain electrons, which require more energy than necessary. In addition, they do not have a strong pull on the valance electrons because they are far away from the nucleus, thus they have less energy for an attraction.



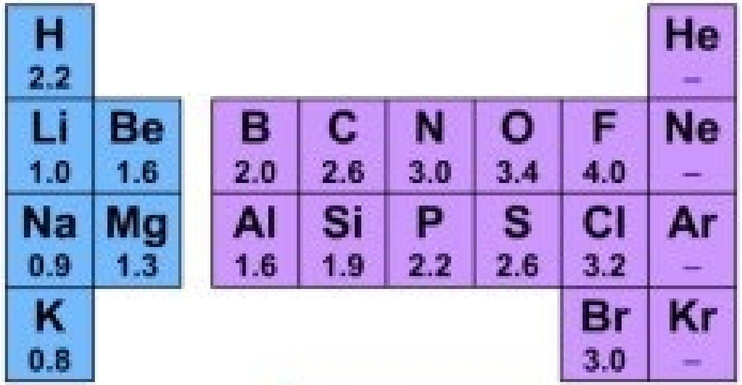
Electronegativity: an atom's ability to attract electrons in a chemical bond.



Electronegativity measures how strongly an atom attracts an electron



Electronegativies increase from bottom left to top right

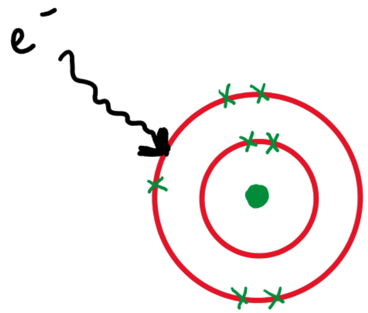


Electronegativity increases across a period because:	atomic radius decreases; nuclear charge increases no extra shielding
Electronegativity decreases down a group because:	atomic radius increases; nuclear charge increases; shielding increases

Comparison EA and Electronegativity

Electron Affinity : Energy released or absorbed by 1 mole of gaseous atom

- Defined in isolated atom



F

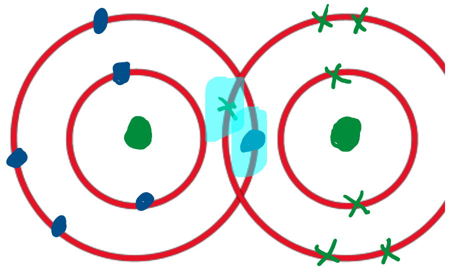
EA = - 328kJ/mol

- In gaseous state
- Has absolute value
- Unit kJ/mol, involved energy
- Experimental values
- EA α $\frac{1}{atom\ size}$ (F<Cl)
- EA α nucleus charge
- EA period L → R increase
  group T → B decrease

Electronegativity: Tendency to attract sherd part

$2s^2\ 2p^2$ $1s^2\ 2s^2$

- Defined in shared or
- bonded state

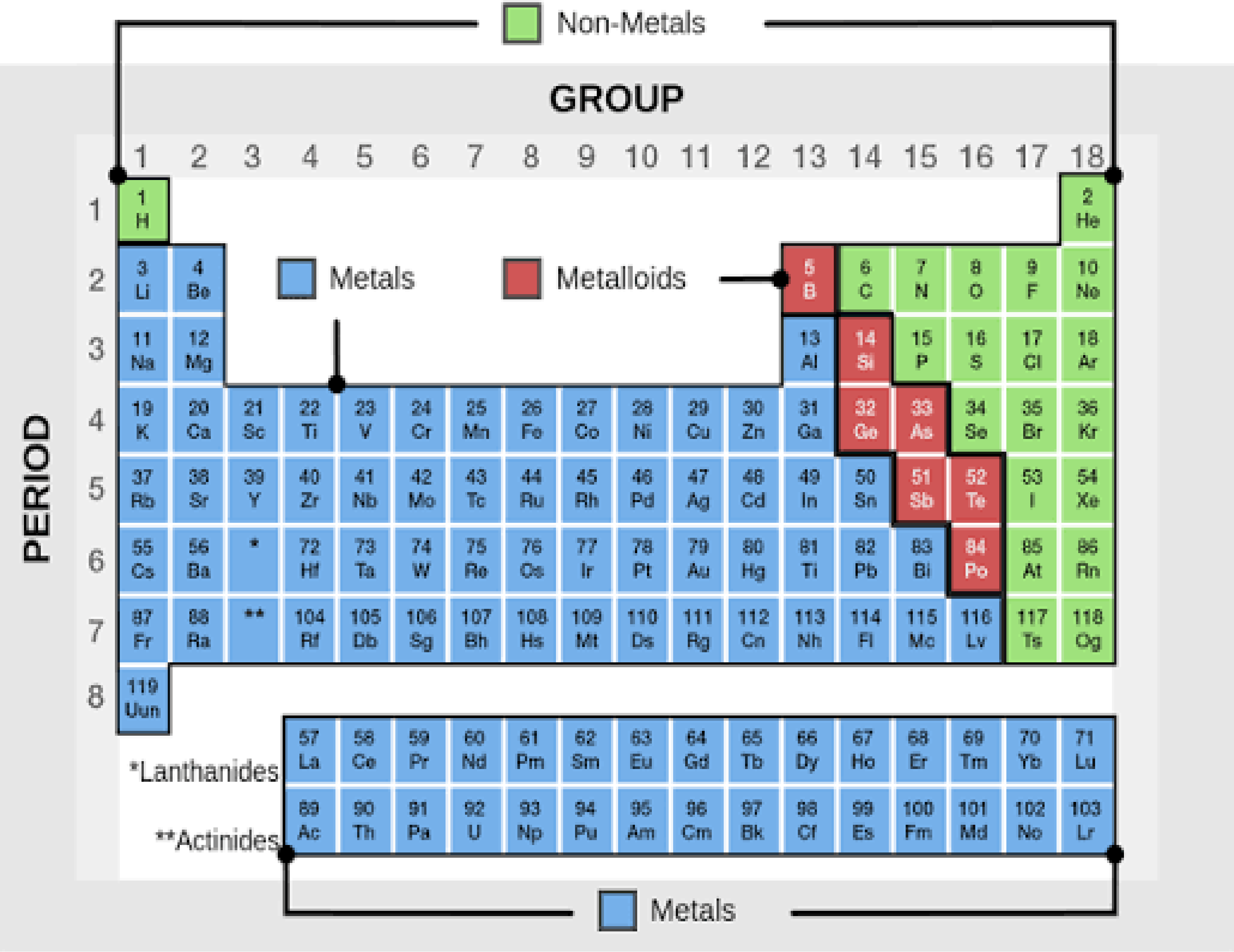


- In all states
- Selective term
  - C – H (C>H)
  - C- Cl (Cl>C)
- No unit (not involve energy)
- Measures on Pauling
- Electronegativity α Zeff
- Electronegativity follow
  diagonal trend period L → R
  Group T → B decreased

PERIODIC TRENDS

PERIODIC PROPERTIES (METALLIC PROPERTIES, BOILING AND MELTING POINT, PERIOD 3 OXIDES)

Portfolio


Non-Metals
GROUP
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
PERIOD
1 2 3 4 5 6 7 8
Metals
Metalloids
1 H
2 He
3 Li 4 Be 5 B 6 C 7 N 8 O 9 F 10 Ne
11 Na 12 Mg 13 Al 14 Si 15 P 16 S 17 Cl 18 Ar
19 K 20 Ca 21 Sc 22 Ti 23 V 24 Cr 25 Mn 26 Fe 27 Co 28 Ni 29 Cu 30 Zn 31 Ga 32 Ge 33 As 34 Se 35 Br 36 Kr
37 Rb 38 Sr 39 Y 40 Zr 41 Nb 42 Mo 43 Tc 44 Ru 45 Rh 46 Pd 47 Ag 48 Cd 49 In 50 Sn 51 Sb 52 Te 53 I 54 Xe
55 Cs 56 Ba * 72 Hf 73 Ta 74 W 75 Re 76 Os 77 Ir 78 Pt 79 Au 80 Hg 81 Tl 82 Pb 83 Bi 84 Po 85 At 86 Rn
87 Fr 88 Ra ** 104 Rf 105 Db 106 Sg 107 Bh 108 Hs 109 Mt 110 Ds 111 Rg 112 Cn 113 Nh 114 Fl 115 Mc 116 Lv 117 Ts 118 Og
119 Uun
*Lanthanides
57 La 58 Ce 59 Pr 60 Nd 61 Pm 62 Sm 63 Eu 64 Gd 65 Tb 66 Dy 67 Ho 68 Er 69 Tm 70 Yb 71 Lu
**Actinides
89 Ac 90 Th 91 Pa 92 U 93 Np 94 Pu 95 Am 96 Cm 97 Bk 98 Cf 99 Es 100 Fm 101 Md 102 No 103 Lr
Metals

Difference Between Boiling Point and Melting Point – **Melting and Boiling**

Temperature at which solid and liquid phases are in equilibrium

Temperature at which the vapour pressure of a liquid is equal to the external pressure.

The **melting or boiling point depends on the intermolecular forces** that exist on two factors:

- 1. The chemical bonding **(ionic, covalent, metallic or van der Waals)**
- 2. The **structure** (the way the atoms and molecules are packed)

The variation of melting and boiling point of elements in the 3rd period is depends on :

Strength of attractive force ↑

- **Metallic structure (Na to Al)**

  Melting point increases →

- **Gigantic covalent structure (Si)**

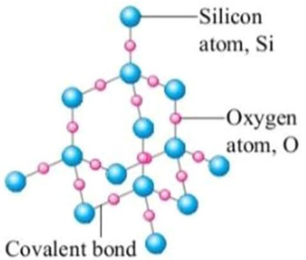
  Held by strong covalent bond, need larger energy to break

- **Simple molecular structure (P to Ar)**

  Ar, Cl_2, P_4, S_8 →

  Melting point increases

  Held by weak van der Waals force, less energy to break

Simple molecule	Covalent compound	Giant molecule
 Water, H_2O  Carbon dioxide, CO_2	Example	 Silicon dioxide, SiO_2
Small and simple structures can be found in the form of solids, liquids or gases.	Structure	Very large structure, usually exists as solids.
Covalent bonds are strong in the molecules and Van der Waals attraction forces between molecules are weak.	Chemical bond	Strong covalent bonds in the molecules only. No Van der Waals attraction forces between molecules because of its giant structure.
Low because only little heat is required to overcome the weak Van der Waals attraction forces between molecules.	Melting point and boiling point	High because a lot of heat is required to break the strong covalent bonds.

Across the period

Melting and boiling point of the 3rd period elements

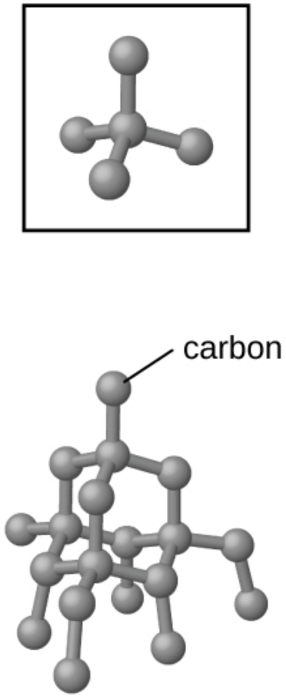
Element	Na	Mg	Al	Si	P	S	Cl	Ar
Melting point (°C)	98	650	660	1410	44	120	-101	-189
Boiling point (°C)	890	1120	2450	2360	280	445	-34	-186

Period	*Group*							
	1	2	13	14	15	16	17	18
Period 2	Li	Be	B	C	N	O	F	Ne
Structure	Giant structure		Giant molecule		Simple molecule			Monatomic
Bonding	Metallic bond		Covalent bond		Covalent bond			–
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
Structure	Giant structure			Giant molecule	Simple molecule			Monatomic
Bonding	Metallic bond			Covalent bond	Covalent bond			–

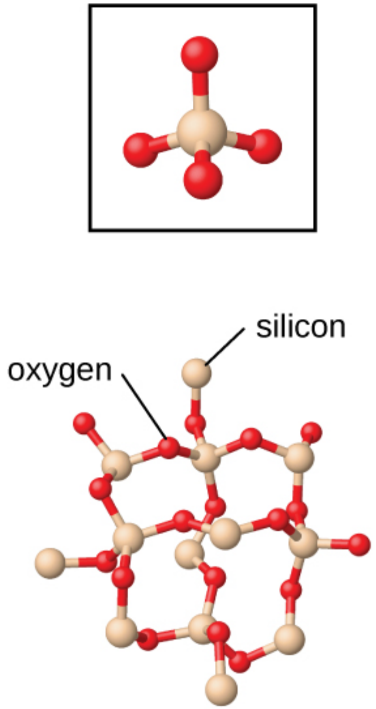
▲ **Table 9.7** Variation in bonding and structure of Periods 2 and 3 elements of the Periodic Table

Gigantic covalent structure (Si)

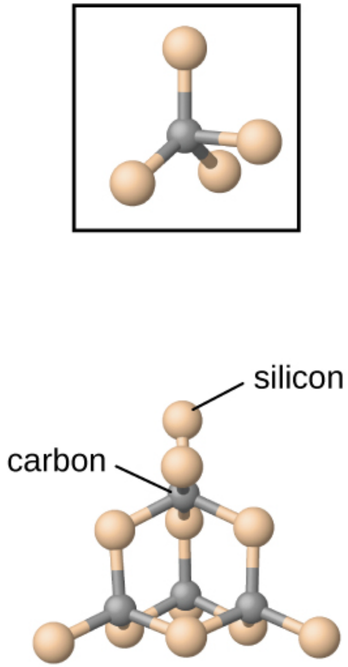
- **Silicon** has a gigantic covalent structure.
- **Melting and boiling point of Si** is **very high** because high energy is needed to break the infinity amount of the strong covalent bond.



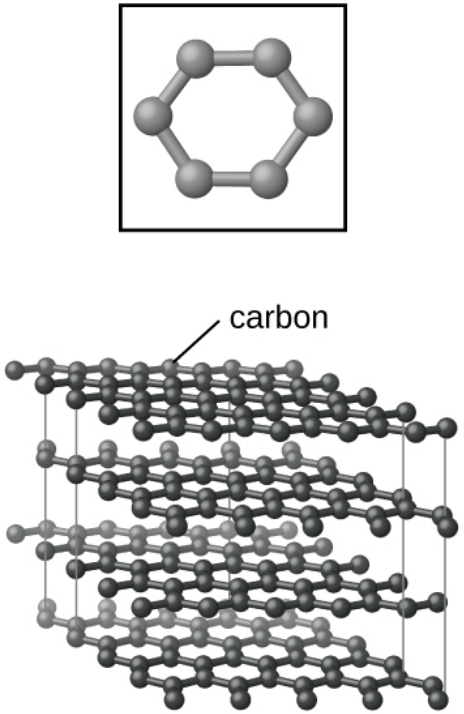
diamond



silicon dioxide



silicon carbide



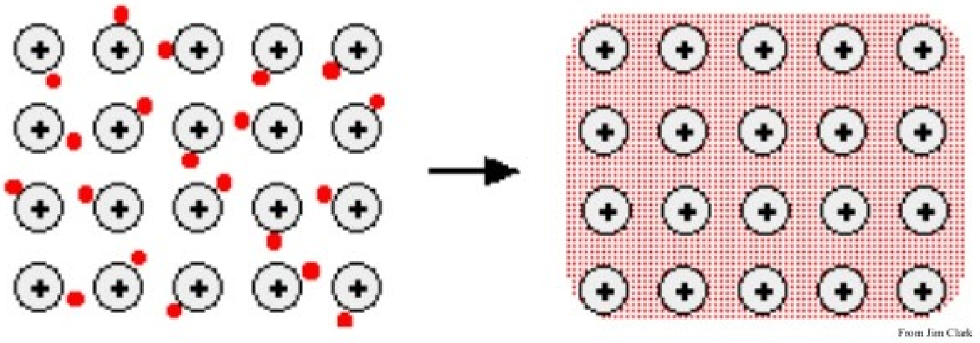
graphite

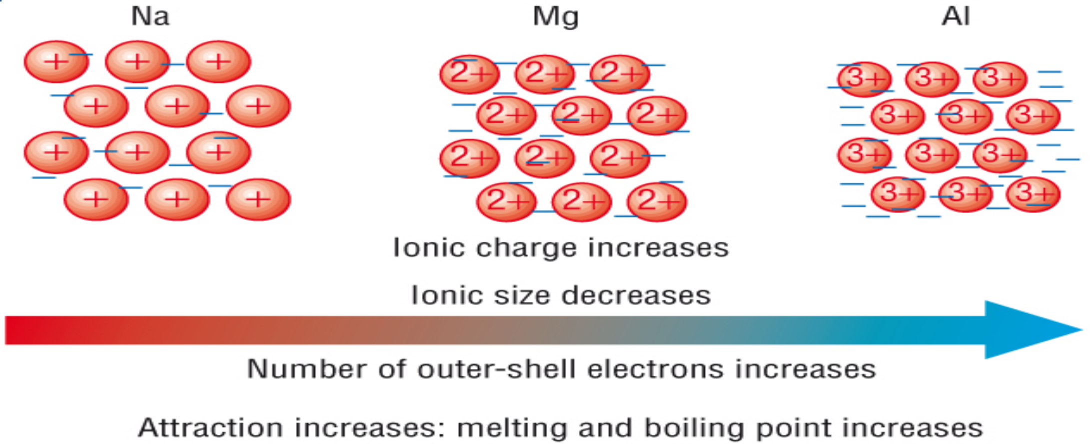
Metallic structure (Na to Al)

- Metal has positive metal ions attracted to the sea of electrons which form the metallic bonding.

- **Strength** of **metallic bonding** is proportional to the number of valence electrons.

The more valence electrons, the stronger the metallic bond and the higher the melting / boiling point





Simple molecular structure (P to Ar)

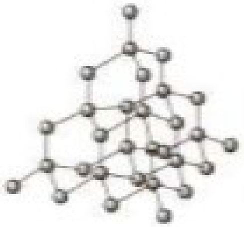
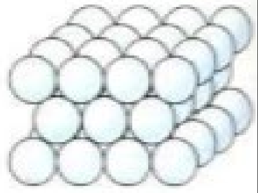
- The non-metal that exist as molecules of P_4, S_8, Cl_2 and Ar (monoatom).

- The **covalent bond between** the **atoms** is very **strong** but the intermolecular force **(Van der Waals), is very weak.**

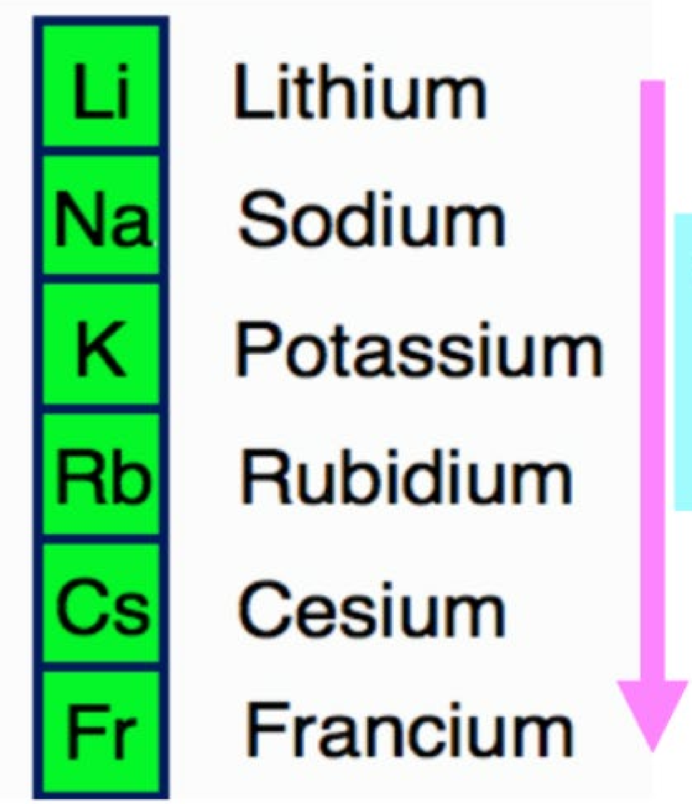
- The strength of Van der Waals force is proportional to molecular size (relative molecular weight)

- Molecular size: $\mathbf{Ar < Cl_2 < P_4 < S_8}$
- Therefore the melting / boiling point : **Ar < Cl < P < S**

Basic Structures and Physical Properties

Bonding	ionic (between metals and non-metals)	covalent (between non-metals)		metallic (between metals)
Structures	giant ionic	giant covalent	simple molecular	giant metallic
Melting point	high	high	low	high
Conduct electricity ?	not when solid, but they do when molten or dissolved in water (when ions are free)	no	no	yes (has free electrons)
Example	sodium chloride 	diamond 	water 	zinc 

Group 1 Element



Physical trends down the group

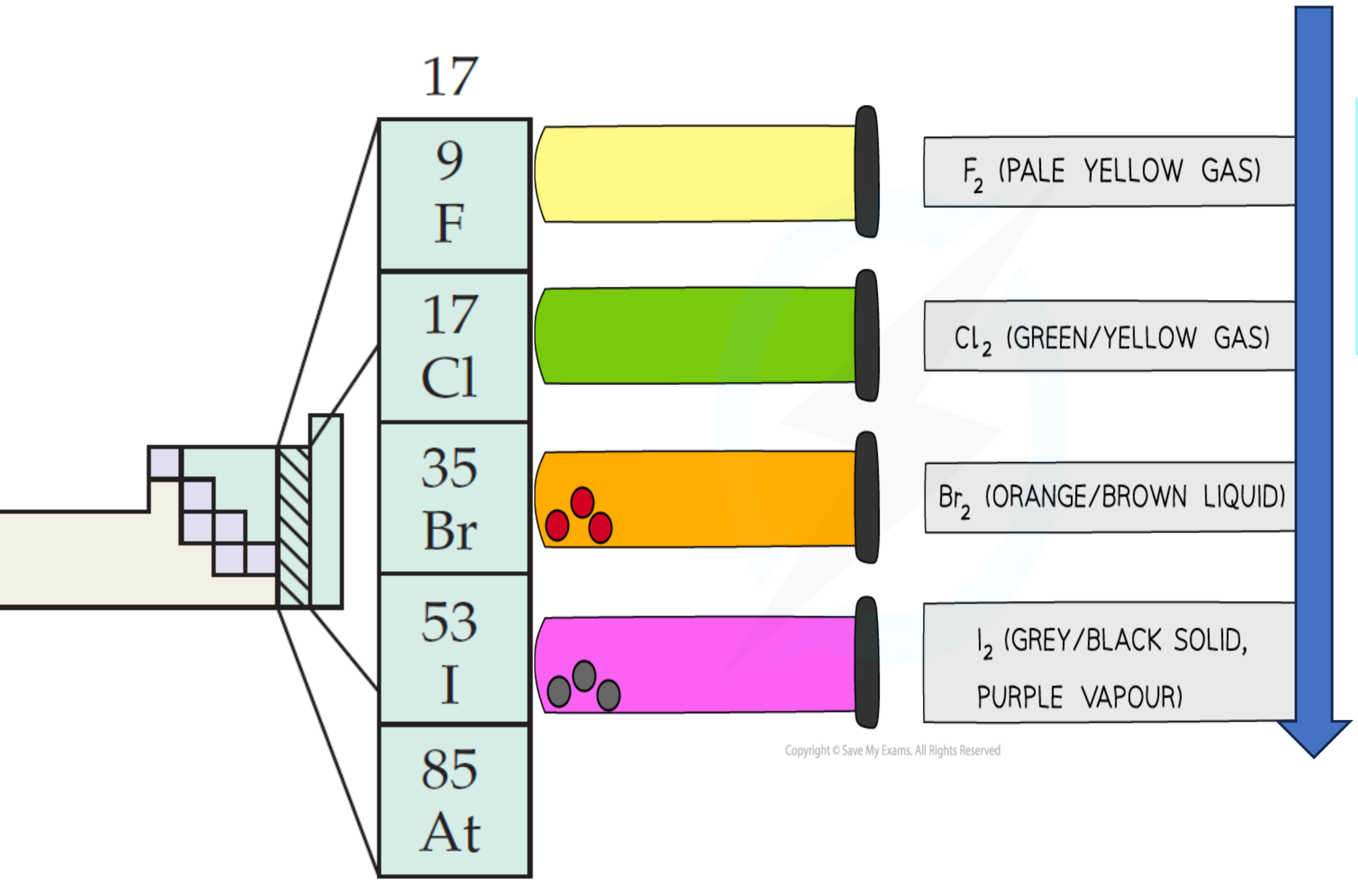
a) Reactivity **increases** down the group.

b) Melting and boiling point **decreases** down the group.

c) Density **increases** down the group.

Element	Melting Point/°C	Boiling Point/°C	Density (g/cm^3)
Lithium	180	1347	0.53
Sodium	98	883	0.97
Potassium	63	774	0.86

Group 17 Element



Physical trends down the group

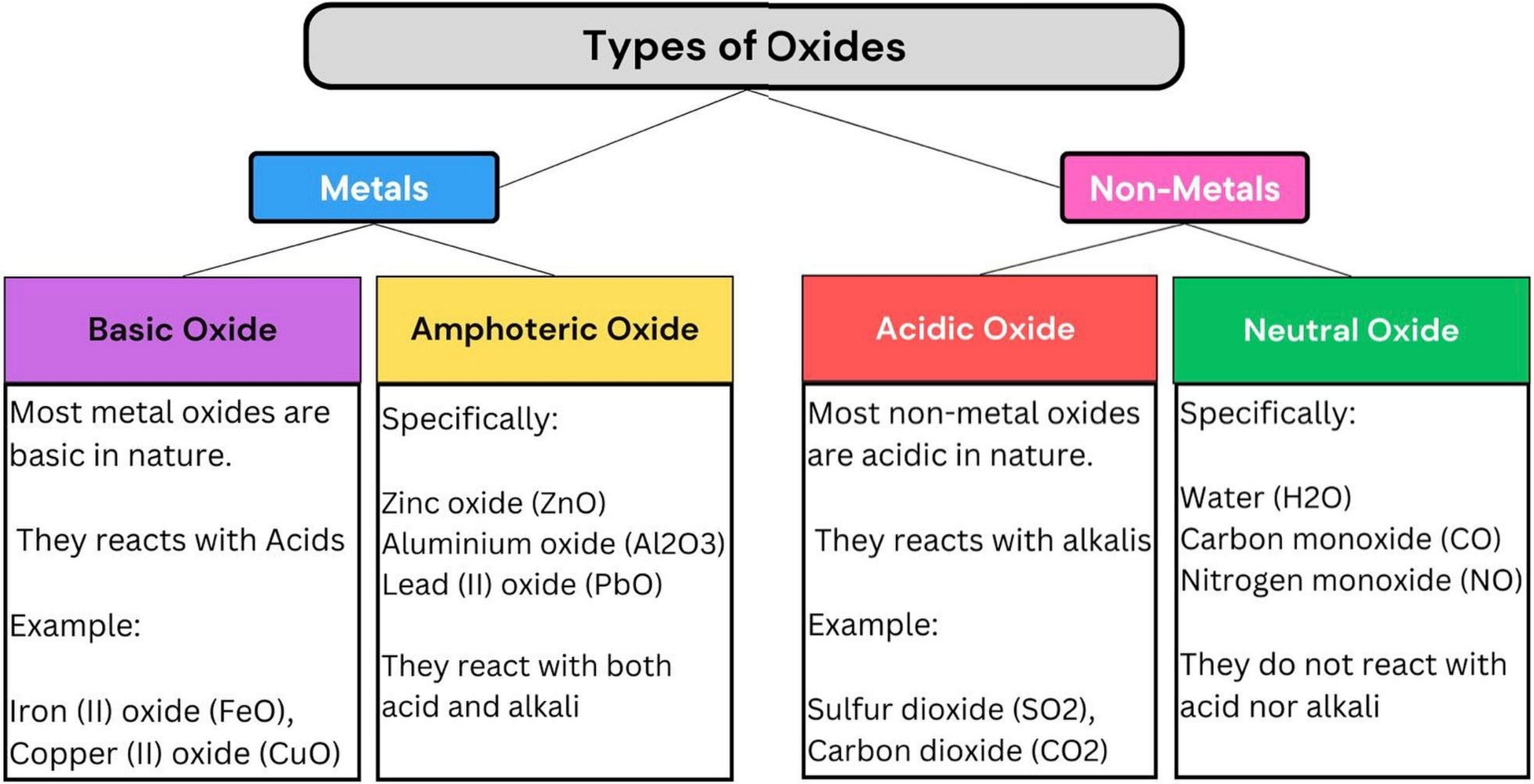
a) Reactivity **decreases** down the group.

b) Melting and boiling point **increases** down the group.

c) Colour becomes **darker** down the group

Element	Melting Point/°C	Boiling Point/°C	Appearance
Chlorine	-101	-34	Greenish-Yellow Gas
Bromine	-7	59	Reddish- Brown Liquid
Iodine	114	184	Purplish-black Solid

Oxides of Period 3 Elements

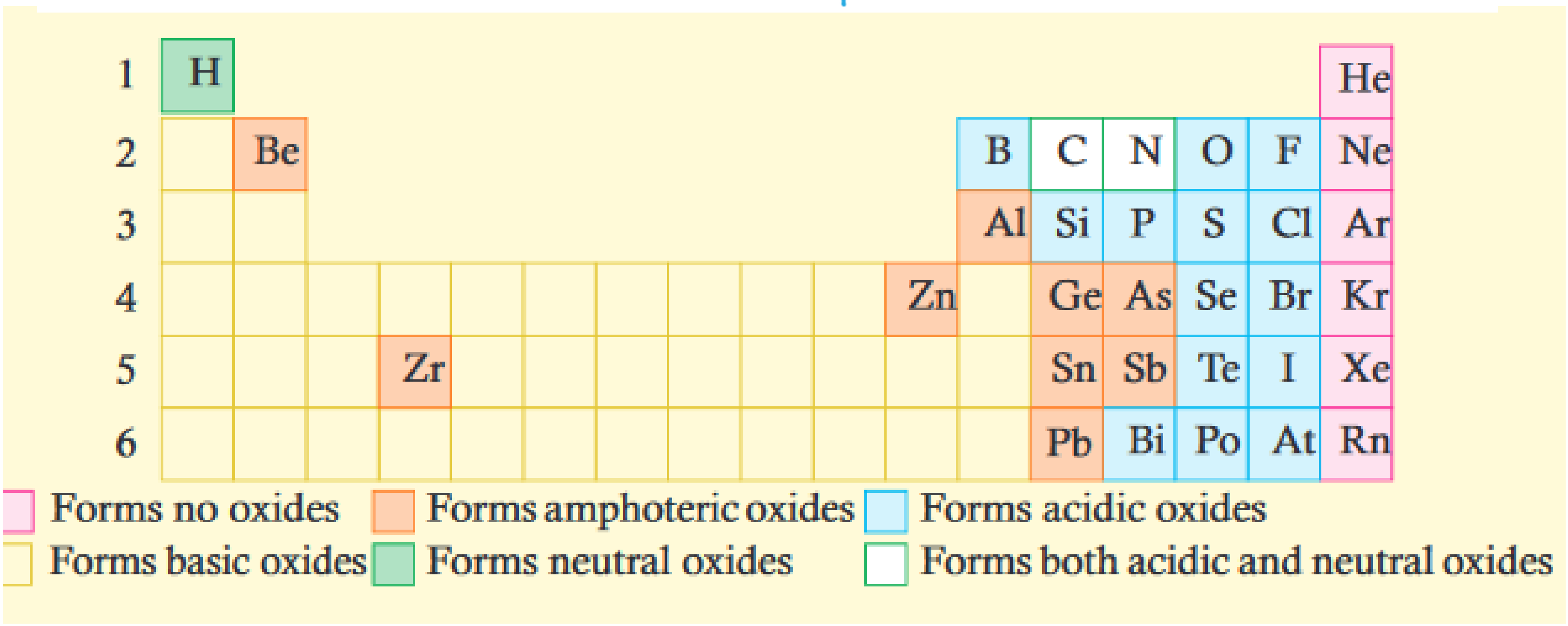

Types of Oxides
Metals
Non-Metals
Basic Oxide
Most metal oxides are basic in nature.
They reacts with Acids
Example:
Iron (II) oxide (FeO),
Copper (II) oxide (CuO)
Amphoteric Oxide
Specifically:
Zinc oxide (ZnO)
Aluminium oxide (Al2O3)
Lead (II) oxide (PbO)
They react with both acid and alkali
Acidic Oxide
Most non-metal oxides are acidic in nature.
They reacts with alkalis
Example:
Sulfur dioxide (SO2),
Carbon dioxide (CO2)
Neutral Oxide
Specifically:
Water (H2O)
Carbon monoxide (CO)
Nitrogen monoxide (NO)
They do not react with acid nor alkali

metals semi-metal non-metal

Na Mg Al Si P S Cl Ar

$3s^1$ $3s^2$ $3s^2\ 3p^1$ $3s^2\ 3p^2$ $3s^2\ 3p^3$ $3s^2\ 3p^4$ $3s^2\ 3p^5$ $3s^2\ 3p^6$

s-block p-block



Period 3 elements when react with **oxygen**:

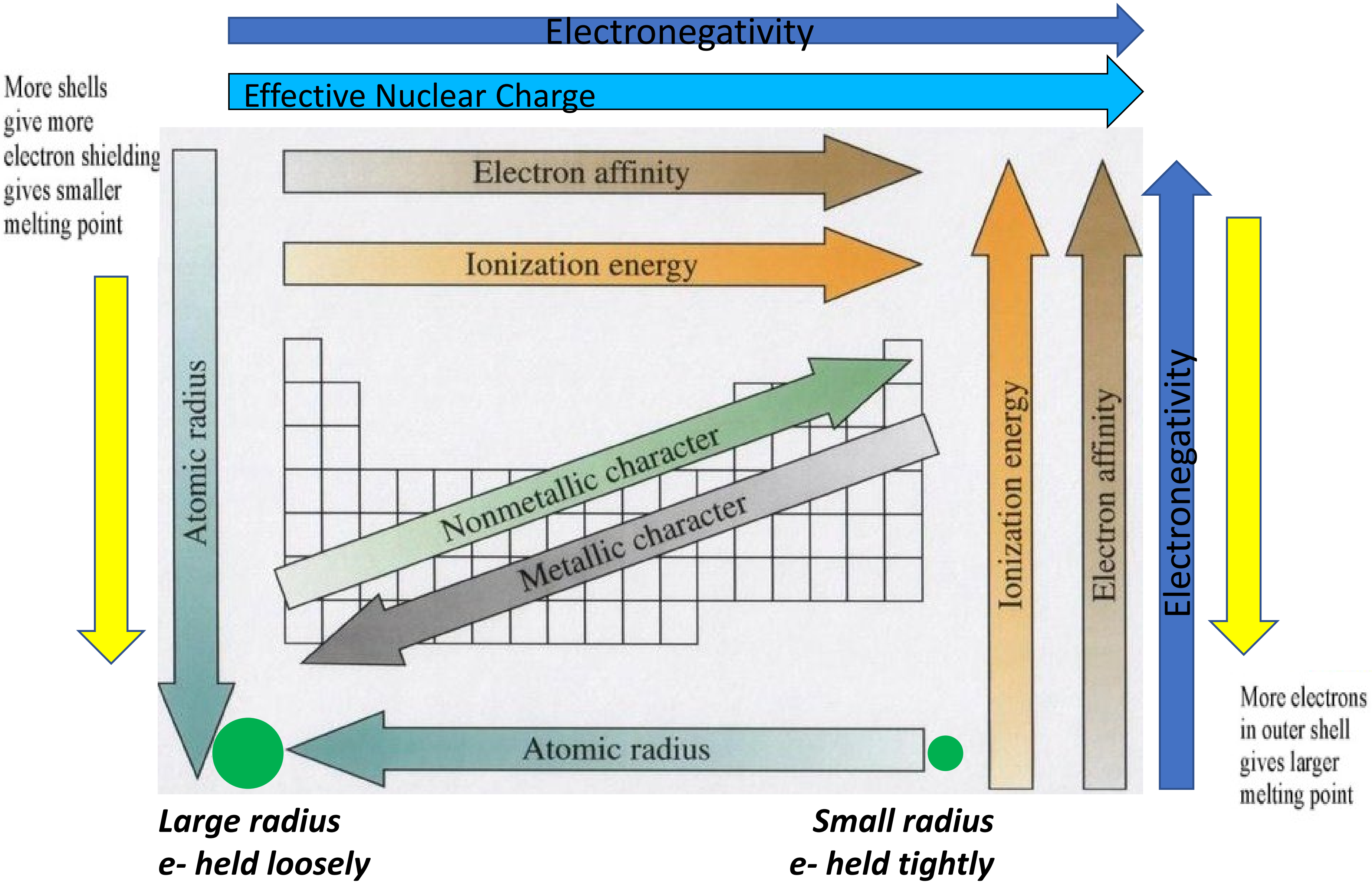
- **Na & Mg** form **basic oxide (most metals)**
- **Al** form **amphoteric oxide (react with both acid and alkali)**
- **Si, P, S & Cl** form **acidic oxide (most non-metals)**

Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2, SO_3
Acid/base nature	Basic	Basic	Amphoteric	Acidic	Acidic	Acidic

Element	Description	Equation
Na	burns vigorously with a yellow flame	$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$
Mg	burns vigorously with a bright white flame	$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$
Al	burns vigorously with a bright white flame	$4Al_{(s)} + 3O_{2(g)} \rightarrow 2Al_2O_{3(s)}$
Si	burns with a bright white flame and white smoke	$Si_{(s)} + O_{2(g)} \rightarrow SiO_{2(s)}$
P	burns spontaneously with a bright white flame and smoke	$4P_{(s)} + 5O_{2(g)} \rightarrow P_4O_{10(s)}$
S	burns with a blue flame	$S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$

OXIDE 3 ELEMENTS	STRUCTURE	BONDING
Na_2O, MgO	The metallic oxides on the left adopt giant structures of ions on the left of the period	The large structures (the metal oxides and silicon dioxide) have high melting and boiling points because a large amount of energy is needed to break the strong bonds (ionic or covalent) operating in three dimensions
Al_2O_3		
SiO_2	Silicon forms a giant covalent oxide (silicon dioxide)	
phosphorus(III) oxide, P_4O_6, and phosphorus(V) oxide, P_4O_{10}	Form simple molecular oxides	The oxides of phosphorus, sulfur and chlorine consist of individual molecules, simple or polymeric.
sulfur dioxide (sulfur(IV) oxide), SO_2, and sulfur trioxide (sulfur(VI) oxide), SO_3		
chlorine(I) oxide, Cl_2O, and chlorine(VII) oxide, Cl_2O_7		

Summary of periodic trends



Questions

1. Based on their positions in the periodic table, predict which has the smallest atomic radius: Mg, Sr, Si, Cl, I.
2. Based on their positions in the periodic table, predict which has the largest first ionization energy: Mg, Ba, B, O, Te.
3. Based on their positions in the periodic table, rank the following atoms in order of increasing first ionization energy: F, Li, N, Rb
4. Atoms of which group in the periodic table have a valence shell electron configuration of ns^2np^3?
5. Based on their positions in the periodic table, list the following atoms in order of increasing radius: Mg, Ca, Rb, Cs.
6. Based on their positions in the periodic table, list the following ions in order of increasing radius: K^+, Ca^{2+}, Al^{3+}, Si^{4+}.
7. Which atom and/or ion is (are) isoelectronic with Br^+: Se^{2+}, Se, As^-, Kr, Ga^{3+}, Cl^-?
8. Compare both the numbers of protons and electrons present in each to rank the following ions in order of increasing radius: As^{3-}, B^{r-}, K^+, Mg^{2+}.
9. Of the five elements Sn, Si, Sb, O, Te, which has the most endothermic reaction? (E represents an atom.) What name is given to the energy for the reaction?

$$E(g) \longrightarrow E^+(g) + e^-$$

10. Which main group atom would be expected to have the lowest second ionization energy?
11. On the basis of their positions in the periodic table, arrange Ge, N, O, Rb, and Zr in order of increasing electronegativity and classify each as a metal, a nonmetal, or a metalloid.